Intermec

Programmer's Reference Manual

ESim for the PC4 and PF8 Printer

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ESim for the PC4 and PF8 Printer

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Document Change Record

This page records changes to this document. The document was originally released as Revision -01.

Version Number	Revision Letter	Date	Description of Change
001	A	3/2008	This document's part number has been revised from P/N 1-960629-03 to P/N 937-011-001 and these commands were added: • USR - Read Serial Number • TD - Define Data Format • TS - Set Real Time Clock
03	A	5/2007	This document has been revised to include the PF8 Bar Code Label Printer.

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Before You Begin

This section provides you with safety information, technical support information, and sources for additional product information.

Safety Information

This section explains how to identify cautions and notes that are in this document.

Caution

A caution alerts you to an operating procedure, practice, condition, or statement that must be strictly observed to prevent equipment damage or destruction, or corruption or loss of data.

Note: Notes either provide extra information about a topic or contain special instructions for handling a particular condition or set of circumstances.

Global Services and Support

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Service	• Get a return authorization number for authorized service center repair. • Request an on-site repair technician.	2 and then choose 1
Service contracts	• Ask about an existing contract. • Renew a contract. • Inquire about repair billing or other service invoicing questions.	1 or 2 and then choose 3

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102-1304 SK Ventium
522 Dangjung-dong
Gunpo-si, Gyeonggi-do Korea, South 435-776
Contact: Mr. Sinbum Kang
Telephone: +82-31-436-1191
E-mail: **sbkang@awoo.co.kr**

IN Information System PTD LTD
6th Floor
Daegu Venture Center Bldg 95
Shinchun 3 Dong
Donggu, Daegu City, Korea
E-mail: **jmyou@idif.co.kr** or **korlim@gw.idif.co.kr**

Who Should Read This Manual

This document is for the person who is responsible for installing, configuring, and maintaining the PC4 or PF8 printer.

This document provides you with information about the features of the ESIM protocol, and how to program bar code labels.

Related Documents

Document Title	Part Number
EasyCoder PC4 Bar Code Label Printer User's Guide	1-960636-xx
EasyCoder PF8d/PF8t Bar Code Label Printer User's Guide	934-011-xxx

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1 Introducing ESim

This chapter introduces the ESim protocol and explains the basic functions and modes of the Esim protocol.

Understanding the ESim Protocol

ESim is a built-in protocol for use with Intermec bar code label printers. ESim allows you to use any computer, terminal, or scanner, or keyboard that can produce ASCII characters to control the printer. ESim is an alternative to Intermec Interdriver, which requires a PC operating in the Microsoft Windows operating system.

With the ESim protocol, you can use any editor to control the printer, using either the serial RS-232 channel, the USB channel, or the parallel Centronics channel.

This manual will assist you in designing labels using the ESim protocol. It has been organized to provide you with an understanding of the printer's functions and command structure.

This manual supports both the PC4 and PF8 printers. Both printers run on different versions of the ESim protocol:

- The EasyCoder PC4 runs on ESim v.6.xx.
- The EasyCoder PF8 runs on ESim v.7.xx.

Note: Any labels created on ESim v.6.xx can be printed on v.7.xx and later, but since new commands are introduced in newer versions of the ESim protocol, labels created on previous versions of ESim cannot be printed.

New commands that are unavailable for the PC4 or ESim v6.25 have been noted in this manual. Other Intermec printers may only support earlier versions of ESim, or may require different firmware. For more information, see your printer user's guide.

Entering a Mode

The printer has 8 modes of operation:

- Autosensing mode
- Dump mode
- Direct mode
- Print mode
- Error mode
- Form Retrieve mode
- Form Edit mode
- Pause Printing mode

When power is first applied to the printer, the printer will look for a startup form named AUTOFR, see **"FS - Form Store" on page 79**. If an AUTOFR form is found, it will be executed in a loop. If no AUTOFR is found, the printer will switch to Direct mode, which is the main operating mode of the ESim protocol.

Autosensing Mode

The Autosensing mode is used to calibrate the sensors that control the media feed, and determines where the individual labels or tags start and stop. There are two different sensors:

- Gap sensor, which detects perforated slots in continuous media, and the gap between self-adhesive labels mounted on a liner (backing paper).
- Black mark sensor, which detects black marks printed on the bottom side of the media.

To enter Autosensing mode

1. Turn off power to the printer, and load it with the appropriate type of media.
2. Switch on power, and hold down the **Feed** key.
3. Select a detection method:
   a. Select gap detection by releasing the **Feed** key within 2 seconds. The indicator lamp will blink green, and the printer will be set to gap detection.

b Select black mark detection by pressing and holding the **Feed** key for more than 2 seconds. The indicator lamp blinks red and the printer is set and calibrated for black mark detection.

4 Once you have selected a detection method, a test label prints, and the printer will then go to Dump mode.

To exit Autosensing mode

- Press the **Feed** key for more than 1 second while autosensing is progress. The printer will exit to Direct mode.

Dump Mode

In Dump mode, the printer will print out the echo of a received ASCII. You can use this mode to debug your software when the printer does not perform as you expect it to.

To enter Dump mode

1 Start up the printer in Autosensing mode. After a test label is printed, the printer will then go to Dump mode. For help, see **"Autosensing Mode" on page 3.**

2 The printer waits for ASCII dump printing. Once you send a string of characters or a label form to the printer, press the **Feed** key. A printout is created.

All characters between ASCII 000 dec. and ASCII 255 dec. received on all communication ports are printed with automatic line-wrapping.

To exit Dump mode

- Press the **Feed** key. A label with the message "out of DUMP" will be printed.

Direct Mode

Direct mode is the main operating mode of the ESim protocol where the printer is set up, graphics and soft fonts are stored, labels and formats are edited, reports are requested, the image buffer can be cleared, the printer can be reset and soft fonts, forms, and graphics can be deleted using various commands.

The Direct mode is also used for retrieving and printing preprogrammed label formats. For a complete list of Direct mode commands go to **"Command List" on page 11**.

The printer will enter Direct mode after you have turned on your printer. Exiting other modes may also switch to Direct mode.

Print Mode

The Print mode is used to print labels, feed out blank media, and control the optional cutter.

To enter Print mode

- In Direct mode, use the P, PA, or C command, or press the **Feed** key. The printer switches to Print mode.

  In Print Pause mode, the printer enters Print mode when the ^C command is executed.

The printer exits Print mode when:

- a printing operation has been successfully performed.
- the printing is paused by command ^A command. For help, see **"^A - Enter Pause Printing Mode" on page 133.**
- the printing is broke by the ^B command. For help, see **"^B - Break Batch Printing" on page 134.**

Error Mode

The Error mode is used for reporting and correcting various types of errors. For more information see **"Error-Handling" on page 153** .

The printer automatically switches to Error mode when an error occurs. After recovery of an error, the indicator lamp switches to green, and the printer returns to the mode it was in previously.

Form Retrieve Mode

The Form Retrieve mode is used for retrieving a form previously edited in Direct mode, and provides its available data fields with data and counter values.

To enter Form Retrieve mode

- In Direct mode, use the FR or ? command. The printer switches to Form Retrieve mode.

The printer exits Form retrieve mode when the number of received lines correspond to the actual number of variables and counters in the form.

Form Edit Mode

This mode is used to permanently store label forms and graphics in the printer memory. In addition to plain text, bar codes, graphics, lines, and boxes, form edit mode also allows the use of variables and counters, which are not available in the Direct mode. The individual label forms can be retrieved and printed in the Direct mode.

Some setup parameters can be included in forms in order to adapt the printer for different applications. However, such setup parameters will affect the global setup after the form has been retrieved and printed.

Pause Printing Mode

The Pause Printing mode is used for pausing or breaking a batch print.

To enter Pause Printing mode

- In Print mode, use the ^A command. Your current print job is paused and the printer switches to Pause Printing mode.

To exit Pause Printing mode

- In Print mode, use the ^C command, or press the Feed key. The printer resumes printing of the job that was paused.

  or

  In the Print mode, use the ^B command. The printer breaks the printing of the paused print job.

Memory

The firmware has memory allocation for print image buffer, form, graphic, and external font memory. The first time the printer is used, it is automatically initialized to default settings, see Chapter 3. Flash memory is used to store information when the power is off. Flash memory chips have a limited lifetime regarding number of writes. It is recommended that all commands that write to Flash memory be used sparingly (that is, not for every label). Commands that write to Flash are Store commands, such as Print Setup, Error Setup, General Setup, Form and Image commands.

Form

Every label is made up of various fields. A form is the complete set of commands that define the content and the design of the label. A form can be saved in memory and retrieved when required.

Text Editor

Use any ASCII output device with a parallel or serial port and a text editor to design the form and programming the printer. Communication is based on the ASCII characters 10 dec. and 32-255 dec.

Commands

The command syntax is based on upper and lower case characters, numeric characters, commas (as separators), quotation marks and line feeds (LF; ASCII 10 dec.). The LF in this manual is listed as ↵ in the end of most command descriptions. Line Feeds (LF) are required to be sent at the end of most command lines. Some commands require to be appended by both a carriage return and a line feed (<CR/LF>).

Note: All programming examples in this manuals start with LF, shown as (↵). It is recommended to start any sequence of command lines with a Line Feed.

Most PC based systems send CR/LF when pressing the **Enter** key. The CR (carriage return) sent in a CR/LF sequence will be ignored. CR alone will not work.

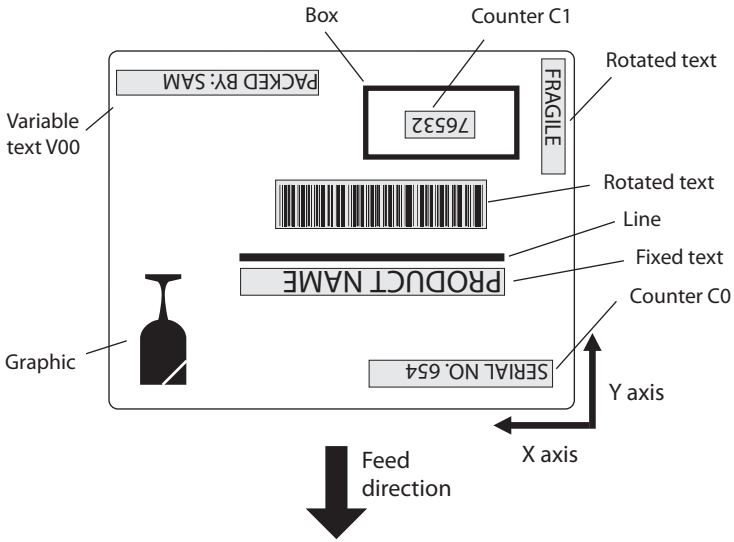
Refer to Chapter 2, **"Command List" on page 11** for information on when the various commands can be used.

Caution

ESim commands and parameters must be used only as described in this manual. Use of unknown commands and parameters may cause permanent printer malfunction.

Field

Each command line of printable data will create a field, which is defined in regard of start position, rotation, magnification, etc.



The illustration shows how a label is printed and fed out when using the default direction.

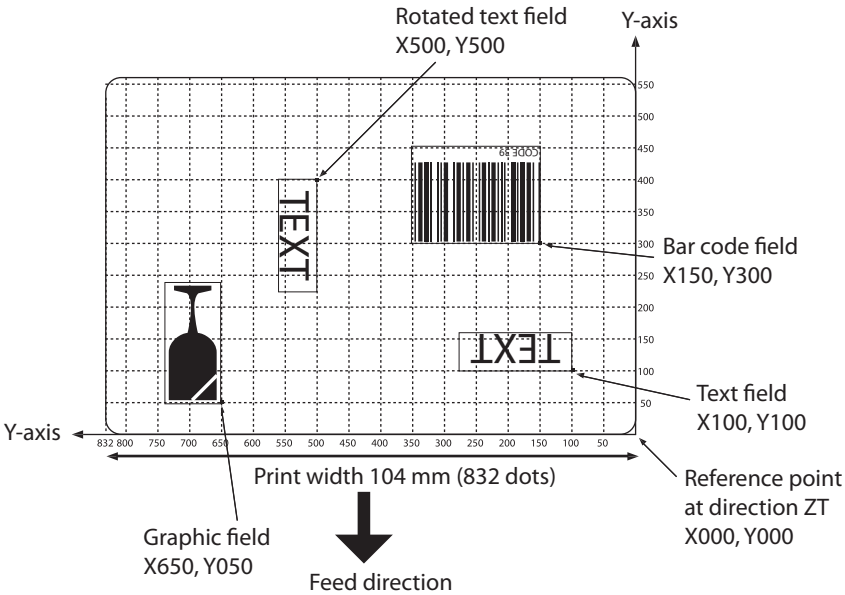
Field Positioning

The printable area of the label forms a grid, where the X-axis runs across the media and the Y-axis runs along the media path. Dots are used as the unit for establishing position of the upper left corner of each field in relation to a specified reference point. The printer is available with two different printhead densities (the 203.2 dpi printer and the 300 dpi printer have different firmware versions).

Printhead Density Specifications

Field	Medium	High
X-axis	203.2 dpi (8 dots/mm)	300 dpi (11.81 dots/mm)
Y-axis	203.2 dpi (8 dots/mm)	304.8 dpi (12 dots/mm)
Number of active dots	832	1208
Maximum print width	104 mm	102.2 mm
Dot line to tear bar	136 dots	204 dots
Dot line to cutter blade	256 dots	384 dots

Text and bar code fields can be rotated around their insertion points, whereas lines, boxes, and graphics cannot be rotated. However, the entire print image can be rotated 180º. The illustration below shows coordinates for the default print direction (ZT).



Example of field positioning when using a 203.2 dpi (8 dots per mm) printhead.

2 Command List

This chapter lists all commands in the ESim protocol.

Commands

The tables below lists all commands available for the ESim protocol for the PC4 and PF8 Bar Code Label Printers.

General Setup Commands

Command	Description
I	Character set selection
i	Asian character spacing
oR	Character substitution
UC	Command reply accept character control
W	Windows mode
Y	Serial port setup

Print Setup Commands

Command	Description
D	Density
f	Cut position
JB	Disable top of form backup
JF	Enable top of form backup
j	Media feed adjustment
O	Option select
Q	Set form length
q	Set label width
R	Set reference point
r	Set relative reference point
S	Speed select
SA	Setup adjust
SPC	Setup print copy
Z	Print direction

Error Setup Commands

Command	Description
eR	User-defined error/status character control
UN	Disable error reporting
US	Enable error reporting

Store and Clean Commands

Command	Description
EK	Delete soft font
ES	Store soft font
GK	Delete graphics
GM	Store graphics in memory
M	Memory allocation

Store and Clean Commands

Command	Description
EK	Delete soft font
ES	Store soft font
GK	Delete graphics
GM	Store graphics in memory
M	Memory allocation

Form Commands

Command	Description
C	Counter
FE	End form store
FK	Delete form
FR	Retrieve form
FS	Form store
PA	Print automatic

Form Commands (continued)

Command	Description
V	Define variable
?	Download variables

Render Commands

Command	Description
A	Print text
B	Print standard bar codes
b	Print two-dimensional bar codes
GG	Print graphics
GRP	Reply graphics from print buffer
GW	Store graphics in image buffer
LE	Line draw exclusive
LO	Line draw black
LS	Line draw diagonal
LW	Line draw white
N	Clear image buffer
X	Draw box

Print Engine Commands

Command	Description
C	Cut immediate
P	Print
PF	Feed media

Information Commands

Command	Description
EI	List soft fonts
FI	Print form information

Information Commands

Command	Description
GI	Print graphics information
U	Print configuration
UDC	Memory cartridge information
UE	Soft fonts information inquiry
UF	Form information inquiry
UG	Graphics information inquiry
UI	Enables prompts and code page inquiry
UM	Code page and memory inquiry
UP	Code page and memory inquiry printing
UV	Product identity and Asian font types

High Priority Commands

Command	Description
^@	Reset printer
^default	Restore factory defaults
^ee	Immediate error report
^A	Enter pause printing mode
^B	Break batch printing
^C	Exit pause printing mode
^D	Remaining labels

3 Setting Up the Printer

This chapter shows the default setup of the printer and gives an example of how to change the setup.

Default Setup

This table shows the default settings on the PC4 or PF8 Bar Code Label Printers.

Default Printer Settings

Parameter	Command	Default Setting
Error/status character	`eR0,0`	XOFF/XON-handling
Density	`D10`	10
Character set	`I8,0,001`	8 bits, code page 0, country code 001
Top of form backup	`JF`	Enabled
Media feed adjust	`J136` `J204`	136 dots at 203.2 dpi 204 dots at 300 dpi
Label gap sensor	`ON`	LTS, TT mode, reverse gap, and cutter disabled.
Character substitution	`oR`	No substitution
Form length	`Q24,0`	Continuous mode, print gap 24 dots
Label width	`q832` `q1208`	832 dots at 203.2 dpi 1208 dots at 300 dpi
Reference point	`R000,000`	X:000, Y:000
Print speed	`S3`	76 mm/sec (3 inches/sec)
Reply character	`UC0`	None
Error handling	`UN`	Error reporting disabled
Windows mode	`WN`	Disabled
Serial port	`Y96,N,8,1`	9600 baud, no parity, 8 data bits, 1 stop bit
Print direction	`ZT`	Start printing at top of image buffer

Printhead Default Settings

Printhead Resolution	Measures
203.2 dpi printheads	203.2 dpi (8 dots/mm) in both directions
300 dpi printheads	300 dpi (11.8 dots/mm) across the media path 304.8 dpi (12 dots/mm) along the media path

Your setup parameters will be reset if:

- the ^default command is used.
- the **Feed** key is pressed more than 3 seconds in dump mode.
- the printer's firmware is upgraded using a firmware cartridge.

Some commands may also affect the values of other commands. For example, if a configuration label is printed (see U and UP commands), the print direction is reset to ZT. Or, if an R reference point command is executed, the label width (see q command) will be changed.

Example

In this example, we will use the PF8 for direct thermal printing. We will print full width thermal top labels in the peel-off mode without using the label taken sensor. The default communication setup and character set are acceptable.

To do this, these setup parameters should be changed in Direct mode:

- Density from 10 to 8
- Media feed adjustment from 136 to 110
- Label taken sensor from enabled to disabled

Enter the following commands:

Command	Explanation
↵	CR/LF to start command structure.
`D8` ↵	Set density.
`j110` ↵	Set media feed adjustment for peel-off operation.
`ON` ↵	Disable label taken sensor.

4 Editing in Direct Mode

This chapter gives you an example of how to create a simple label in Direct mode.

Example

In this example, we will print two copies of a label which we will edit in Direct mode. The label will look like the example in Chapter 5.

The label can be printed as many times as you want, as long as it still is stored in the image buffer. Once replaced, it cannot be retrieved. It also implies that counters and variables cannot be used.

Before entering commands, you must do the following:

- Make sure that the printer has been set up for the application.
- Make sure that the length of the label and the gap has been determined by printing a test label. For more information on how to print a test label, see **"Autosensing Mode" on page 3**.
- Verify that the graphic used in the example has been downloaded to the printer, see the GM command.

Command	Explanation
↵	CR/LF to start command structure.
`N` ↵	Clear image memory.
`X0,0,4,752,584` ↵	Draw a box.
`LO0,144,752,4` ↵	Disable label taken sensor.
`A40,400,1,1,1,1,N,"Made in Sweden"` ↵	Write a 90° text line of fixed data.
`A24,160,0,5,1,1,R,"EASYCODER"` ↵	Write a text line of fixed data.
`A24,250,0,4,1,1,N,"MODEL: 501SA"` ↵	Write a text line of fixed data.
`A472,312,0,4,1,1,N,"Checked by: Dan"` ↵	Write a text line of fixed data.
`A24,312,0,4,1,1,N,"SERIAL#: 000001"` ↵	Write a text line of fixed data.
`B280,440,0,1,2,3,96,B,"S00000"` ↵	Write a text line of fixed data.
`GG24,12,"LOGO"` ↵	Write a graphic from graphics memory.
`P1` ↵	Print command to image buffer, and prints 1 copy.

Note: The Intermec logotype is not included in the software package. For this example, you can use any graphic with approximately the same size. You can also omit the GG command from the example until you are more familiar with it.

5 Editing in Direct Mode

This chapter gives you an example of how to create a simple label using a form.

Example

In this example, we will edit a label that can be saved as a form in the printer's memory and retrieved when so required. It also means that we can use counters and variables.

Before entering commands, you must do the following:

- Make sure that the printer has been set up for the application.
- Make sure that the length of the label and the gap has been determined by printing a test label. For more information on how to print a test label, see **"Autosensing Mode" on page 3**.

Verify that the graphic used in the example has been downloaded to the printer, see **"GM - Store Graphics In Memory" on page 82**.

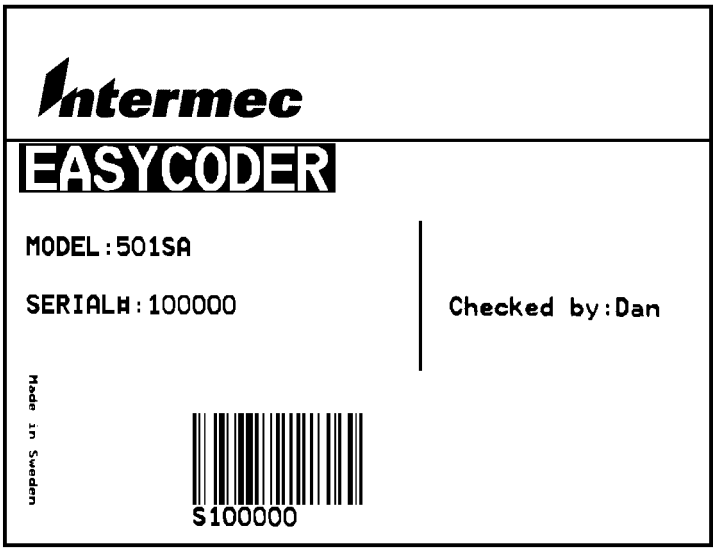


Illustration of the finished label.

Note: The Intermec logotype is not included in the software package. For this example, you can use any graphic with approximately the same size. You can also omit the GG command from the example until you are more familiar with it.

Name the Form

Name of this form is TEST.

Command	Explanation
↵	CR/LF to start command structure.
`FK"TEST"` ↵	Delete any existing form named TEST.
`FS"TEST"` ↵	Start store form named TEST.

Define Variables

The first variable (V00) has a maximum size of 15 characters.
The second variable (V01) has 10 characters and prints in reverse.
The third variable (V02) has a maximum size of 8 characters.

Command	Explanation
`V00,15,N,"Enter Product name:"` ↵	Define first variable.
`V01,10,L,"Enter Model number:"` ↵	Define second variable.
`V02,8,N,"Checked by":"` ↵	Define third variable.

Note: The text within the quotes are prompts, which will be sent from the printer to host when the label form is retrieved (serial communications only).

Define a Counter

The counter has a maximum 6 digits.

Command	Explanation
`C0,6,L,+1,"Enter Serial Number:"` ↵	Define counter.

Note: The variables V00, V01, V02, and the counter C0 are defined within the label form named TEST. The next label form containing variables and counters will start again with V00 and C0. If variable data is being sent from an external database, omit the text between the quotes and replace with a space character, for example V00,15,N,“ ”.

Draw a Box and Two Lines

Start to draw the surrounding box using the X command and then draw the two lines using the LO command.

Command	Explanation
`X0,0,4,752,584` ↵	Draw a box.
`LO0,144,752,4` ↵	Draw a line.
`LO440,232,4,160` ↵	Draw a line.

Place a Text Line With Fixed Data

Enter a 90° rotated text line containing the fixed data “Made in Sweden” in text size 1. The quotation marks enclosing the fixed data will not be printed. The text size (1) is the smallest resident font in the printer.

Command	Explanation
`A40,400,1,1,1,1,N,"Made in Sweden"` ↵	90° text line containing fixed data.

Place a Variable Text

The next line is a text line, using text size 5 in reverse and prints the variable V00. The data printed in this field must be sent to the printer at the time of form retrieval.

Command	Explanation
`A24,160,0,5,1,1,R,V00` ↵	Write a text line, first variable.

Place a Combination of Fixed Data and a Variable

The following two command lines consist of a combination of fixed data enclosed in quotation marks and variable data.

Command	Explanation
`A24,250,0,4,1,1,N,"MODEL:"V01` ↵	Write a text line, second variable.
`A472,312,0,4,1,1,N,"Checked by:"V02` ↵	Write a text line, third variable.

Place a Combination of Fixed Data and a Counter

The next command line is a text line containing fixed data and the counter (C0). The first time this label form is retrieved for printing, the start value for this counter must be sent to the printer. The printer will store the value of the counter for this form and automatically continue to print the next value the next time this form is retrieved. Reset or set to another value by sending a new start value.

Note: The value of the counter will be kept in the memory even if another form is retrieved or the printer is switched off.

Command	Explanation
`A24,312,0,4,1,1,N,"SERIAL#:"C0"` ↵	Write a text line with fixed data and first counter.

Place a Bar Code with Fixed Data and a Counter

Below a Bar Code Command line is used to enter a Code 128 bar code, containing the fixed data “S” in combination with the actual counter value. It is also set for printing the human readable text below the bar code.

Note: The narrow to wide ratio is not relevant for Code 128. The printer will use the value for the narrow bar to define the bar code (value 3 for wide bar definition is ignored).

Command	Explanation
`B280,440,0,1,2,3,96,B,“S”C0` ↵	Bar code with fixed data and first counter.

Place Graphics

The next line writes a graphic named “Intermec” from memory and positions it on the form.

Command	Explanation
`HH24,12,“LOGO”` ↵	Write graphic from graphics memory.

End Programming of This Form

The closing command flags the end of form, see the full program listing in the next section.

Command	Explanation
`FE` ↵	Closing command to store form.

Complete List of the Example

Command	Explanation
↵	CR/LF to start command structure.
FK"TEST" ↵	Delete any existing form named TEST.
FS"TEST"	Start store form named TEST.
V00,15,N,"Enter Product name:" ↵	Define first variable.
V01,10,L,"Enter Model number:" ↵	Define second variable.
V02,8,N,"Checked by":" ↵	Define third variable.
C0,6,L,+1,"Enter Serial Number:" ↵	Define counter.
X0,0,4,752,584 ↵	Draw a box.
LO0,144,752,4 ↵	Draw a line.
LO440,232,4,160 ↵	Draw a line.
A40,400,1,1,1,1,N,"Made in Sweden" ↵	90° text line containing fixed data.
A24,160,0,5,1,1,R,V00 ↵	Write a text line, first variable.
A24,250,0,4,1,1,N,"MODEL:"V01 ↵	Write a text line, second variable.
A472,312,0,4,1,1,N,"Checked by:"V02 ↵	Write a text line, third variable.
A24,312,0,4,1,1,N,"SERIAL#:"C0" ↵	Write a text line with fixed data and first counter.
B280,440,0,1,2,3,96,B,"S"C0 ↵	Bar code with fixed data and first counter.
HH24,12,"LOGO" ↵	Write graphic from graphics memory.
FE ↵	Closing command to store form.

6 Retrieving and Printing a Form

This chapter gives a step-by-step example of how a form is retrieved. The variable data fields are manually provided with information after the operator has been prompted to do so on the screen of the host. Finally, the operator orders a label to be printed.

Example

The form “TEST”, edited in the previous chapter can be retrieved and printed form any ASCII sending device using this sequence:

Command	Explanation
↵	CR/LF to start command structure
`FR“TEST”` ↵	Retrieve form
`?` ↵	Call for variables
`EASYCODER` ↵	Substitute variable V00
`501SA` ↵	Substitute variable V01
`Dan` ↵	Substitute variable V02
`100000` ↵	Counter start value C0
`P1,2` ↵	Print 2 copies of a single label

In this example, we have manually substituted variables for testing purposes.

Note: It is critical to the syntax to send exactly the same number of variable lines as defined for this label form.

Provided you use the serial interface for communication between printer and host, you can make the printer return prompts that appear on the screen, requesting the operator to enter input data, by sending a UI command after each power-up. The optional Keyboard Display Unit automatically sends the UI command at power-up.

Printer Prompt	Command	Explanation
	↵	CR/LF to start command structure
	`UI`	Enables prompts command (optional)
UI80,001		Printer returns code page status
	`FR "TEST"` ↵	Retrieve form
	`?` ↵	Call for variables
Enter Product name:	`EASYCODER` ↵	Substitute variable V001
Enter Model number:	`501SA` ↵	Substitute variable V01
100001	`100000` ↵	Reset, accept, or enter 2 counter start value C0
Number of label sets		Prompt
P1		Ignore
	`P1` ↵	Enter P and quantity of labels
Copies of each label		Prompt
1	`2` ↵	Enter number of copies ↵

The example below demonstrates that it is not necessary to set the counter start value again. The counter internally keeps track of the last number issued as long as the power remains switched on and is updated according to instructions in the form.

Command	Explanation
↵	CR/LF to start command structure
`FR"TEST"` ↵	Retrieve form
`?` ↵	Call for variables
`EASYCODER` ↵	Substitute variable V00
`501SA` ↵	Substitute variable V01
`Dan` ↵	Substitute variable V02
↵	CR/LF to use existing counter value
`P1,2` ↵	Print 2 copies of a single label

Once a form has been retrieved, it can be used over and over again until another form is retrieved. All variable input data and counter values are stored in the volatile memory, which means they will be lost if the printer is switched off or at a power failure. If prompts are enabled, existing data and counter values will be displayed on the screen after the related prompt. Any input data can be overwritten at will.

Command	Explanation
`?` ↵	Call for variables in the same form
↵	CR/LF to use existing data in V00
↵	CR/LF to use existing data in V01
`Sam` ↵	Substitute data in variable V02
`200000` ↵	Substitute counter start value
`P1,1` ↵	Print 1 copy of 1 label

Note: the question mark (?) following the FR command is essential for the printing of certain fields edited in Form Edit mode, that is fields containing variables or counters. Variables and counter start values must be entered or accepted as described above. If no question mark is transmitted, all fields containing variable input, that is variables and counters, will be completely omitted from the printout.

7 Commands

This chapter lists the various commands in alphabetic order. For each command, a short description is given, followed by the syntax for the command, an explanation of the parameters included in the syntax, and an example of how to use the commands are given.

ESIM Commands

In the command syntax, there are a few conventions for substituting data or indicating how data can be used:

- p_1 - p_n indicates parameters listed separately below the command syntax.
- Square brackets [ ] indicate optional parameters or data.
- Vertical lines | indicate alternatives
- “Name” indicates you to enter the name of a form or graphic within the quotation marks (ASCII 34 dec.), for example “Intermec”.
- “Data” indicates that the data could be from another source such as a .PCX file, a database, or entered by the operator. “Data” designates the place in the command sequence to input data.

  Because firmware uses “ ” (ASCII 34 dec.), you need a special designator if you need to print text or bar codes that include these quotation marks. The backslash character \ (ASCII 92 dec.) serves that purpose. Examples are shown in the table below.

Printing Examples

To print	Enter
“	`"\""`
“ABC”	`"\"ABC\""`
\	`"\\"`
\code\	`"\\ccde\\"`

Note: If a 7 bit character set is selected, the quotation marks “ ” will not be supported, and all backslash \ characters will be printed as entered.

- A trailing arrow → indicates that data continues on the next line.
- A leading arrow ← indicates that data continued from the previous line.

- Brackets < > is an alternative method for writing unprintable ASCII characters, for example <ACK> = ASCII 06 dec.

A - Printing Text

Purpose: The A command is used to print an ASCII text string.

Syntax: Ap_1, p_2, p_3, p_4, p_5, p_6, p_7,"DATA"

Parameter Descriptions

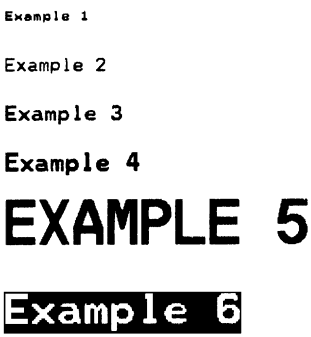
Parameter	Description	
p_1	Horizontal start position (x) in dots.	
p_2	Vertical start position (y) in dots.	
p_3	Rotation:	
	0 = No rotation (left to right)	
	1 = 90° rotation (left to right)	
	2 = 180° rotation (left to right)	
	3 = 270° rotation (left to right)	
	4 = No rotation. Top to bottom. Asian fonts (p4=8) only.	
	5 = 90° rotation. Top to bottom. Asian fonts (p4=8) only.	
	6 = 180° rotation. Top to bottom. Asian fonts (p4=8) only.	
	7 = 270° rotation. Top to bottom. Asian fonts (p4=8) only.	
p_4	Font selection:	
	203.2 dpi (8 dots/mm)	**300 dpi (11.81 dots/mm)**
	1 = 203.3 cpi, 6 points (8 x 12 dots)	25 cpi, 4 points (12 x 20 dots)
	2 = 16.9 cpi, 7 points (10 x 16 dots)	8.75 cpi, 6 points (16 x 28 dots)
	3 = 14.5 cpi, 10 points (12 x 20 dots)	15 cpi, 8 points (20 x 36 dots)
	4 = 12.7 cpi, 12 points (14 x 24 dots)	12.5 cpi, 10 points (24 x 44 dots)
	5 = 5.6 cpi, 24 points (32 x 48 dots)	6.25 cpi, 21 points (48 x 80 dots)
	8 = Asian fonts:	
	Korean, Korea.24	(24 x 24 dots in a double-byte)
	Chinese GB, GB.24	(24 x 24 dots in a double-byte)
	Chinese BIG-5, Big5.24	(24 x 24 dots in a double-byte)
	Japanese, Japanese.24	(24 x 24 dots in a double-byte)
	Latin	(24 x 24 dots in a single-byte, included in all .24 fonts) (dot size controlled by ES command.)

Parameter Descriptions (continued)

Parameter	Description
p5	Horizontal multiplier 1, 2, 3, 4, 5, 6, 7, 8
p6	Vertical multiplier 1, 2, 3, 4, 5, 6, 7, 8, 9
p7	N = Normal image R = Reverse image
"DATA"	Represents a fixed data field. When using Asian double-byte fonts, specify both bytes as ASCII decimal values, starting with the first value being larger than ASCII 127 dec (ASCII 7F hex), or below ASCII 127 (ASCII 7F hex) decimal to specify a Latin character.

Example:

```
↵
N ↵
A50,0,0,1,1,1,N,"Example 1" ↵
A50,50,0,2,1,1,N,"Example 2" ↵
A50,100,0,3,1,1,N,"Example 3" ↵
A50,150,0,4,1,1,N,"Example 4" ↵
A50,200,0,5,1,1,N,"EXAMPLE 5" ↵
A50,300,0,3,2,2,R,"Example 6" ↵
P1 ↵
```



Note: Font size 5 only supports uppercase characters, see example 5 in the label above.

Remarks: The "DATA" field can be replaced by or combined with the commands below:

DATA Field Command Descriptions

Command	Description
Variable:	
Vnn	Prints the contents of the variable "nn" at this position, where n is a 2 digit number from 00-99.
Consecutive number counter:	
Cn	Prints the contents of counter "n" at this position.
Cn±x	Prints the contents of counter "n" at this position while setting the counter's start value to "x". n and x are 1 digit numbers from 0-9. Enter + to increase the value or - to decrease the value.

Example 1:

The command Cn±x in this example will be used twice and count up the single counter by one (1) in each position (last two A-command lines).

Set the Form Step Value p4 to +3 for the counter Cn used in our example (see the C-command line) and refer to the C-Counter command.

```
↵
FK"TEST2" ↵
FS"TEST2" ↵
C0,5,L,+3,"Counter 0" ↵
A180,50,0,3,1,1,N,C0 ↵
A380,50,0,3,1,1,N,C0+1 ↵
A580,50,0,3,1,1,N,C0+2 ↵
FE ↵
```

Example 2:

This example illustrates how fixed text, variable text, and counters can be used in text fields in Form Edit mode:

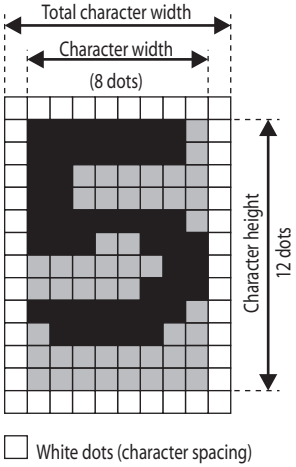
```
↵
FK"TEST1" ↵
FS"TEST1" ↵
V00,25,1,"Product name" ↵
C0,4,L,+1,"Start serial No" ↵
A50,50,0,4,1,1,N,"COMPANY NAME"  ↵
A50,100,0,3,1,1,N,"Product: V00 ↵
A50,150,0,3,1,1,N,"Serial No: "C0
FE ↵
```

Combination of several options can be used in a single text field:

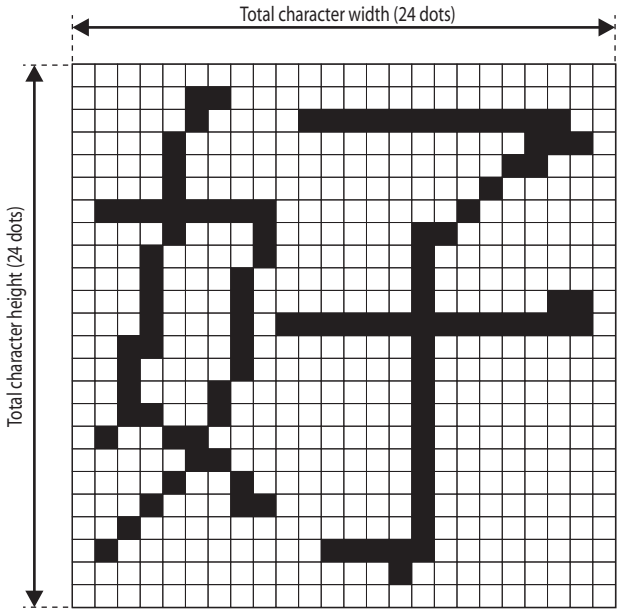
```
A50,300,0,3,2,2,R,"Deluxe"V01C1"Combo" ↵
```

This example writes the text "Deluxe", the contents of variable 01, the contents of counter 2, the text "Combo", and the contents of variable 01.

The Latin font characters (1–5) are dot-mapped differently than the Asian font characters (8). Asian characters do not have any built-in gap between individual characters and will thus be placed adjacently, whereas Latin characters include a single dot border around each character. To create inter-character spacing for Asian fonts, use the i command.



Latin Characters



Asian Characters

The Asian fonts can print character strings running from top to bottom (p_3 = 4–7), as well as the standard Latin word orientation from left to right (p_3 = 0–3). The characters will print in the sequence that they are entered into the data field of the A command.

B - Standard Bar Codes

Purpose: This command is used to print standard bar codes.

Syntax: Ap_1, p_2, p_3, p_4, p_5, p_6, p_7, p_8, "DATA"

Parameter Descriptions

Parameter	Description
p_1	Horizontal start position (x) in dots.
p_2	Vertical start position (y) in dots.
p_3	Rotation: 0 = No rotation 1 = 90° rotation clockwise 2 = 180° rotation clockwise 3 = 270° rotation clockwise
p_4	Bar code select. See bar code on next page.

Parameter Descriptions (continued)

Parameter	Description		
p5	**Bar Code Type:**	"p4"	"p5"
	Code 39 standard or extended	3	1-10
	Code 39 with check digit	3C	1-10
	Code 93	9	1-10
	Code 128 UCC case code	0	1-10
	Code 128		
	Automatic subset selection	1	1-10
	Manual selection of subset A	1A	1-10
	Manual selection of subset B	1B	1-10
	Manual selection of subset C	1C	1-10
	Codabar	K	1-10
	EAN8	E80	2-4
	EAN8 2 digit add-on	E82	2-4
	EAN8 5 digit add-on	E85	2-4
	EAN13	E30	2-4
	EAN13 2 digit add-on	E32	2-4
	EAN13 5 digit add-on	E35	2-4
	Interleaved 2 of 5	2	1-10
	Interleaved 2 of 5 with check digit	2C	1-10
	Interleaved 2 of 5 with human readable check digit	2D	1-10
	Postnet 5, 6, 8, and 11 digit	P	n.a
	Planet	PL	n.a
	UCC/EAN 128	1E	1-10
	UPC A	UA0	2-4
	UPC A 2 digit add-on	UA2	2-4
	UPC A 5 digit add-on	UA5	2-4
	UPC E	UE0	2-4
	UPC E 2 digit add-on	UE2	2-4
	UPC E 5 digit-add-on	UE5	2-4
	UPC SCC	2U	1-10
p6	Wide bar width in dots (2-30).		
p7	Bar code height in dots.		
p8	B = Human readables ON N = Human readables OFF.		

Parameter Descriptions (continued)

Parameter	Description
"DATA"	Represents a fixed data field. If empty, no part of the bar code will be generated. i.e no start nor stop bars.

Example:

This example produces a Code 39 bar code:

```
↵
N ↵
B50,50,0,3,2,6,200,B,"998152-001" ↵
P1 ↵
```

Remarks: In all bar codes, the "DATA" field can be replaced or combined with the commands below:

DATA Field Command Descriptions

Command	Description
Variable:	
Vnn	Prints the contents of the variable "nn" at this position, where n is a 2 digit number from 00-99.
Consecutive number counter:	
Cn	Prints the contents of counter "n" at this position.
Cn±x	Prints the contents of counter "n" at this position while setting the counter's start value to "x". n and x are 1 digit numbers from 0-9. Enter + to increase the value or - to decrease the value.

Example 1:

When labels with consecutive numbers are printed next to each other across the web, it is done by using a single counter in a single form.

The command Cn±x in our example will be used twice and count up the single counter by one (1) in each position (last two B-command lines).

Set the Form Step Value p4 to +3 for the counter Cn used in our example (see the C-command line) and refer to the C-Counter command.

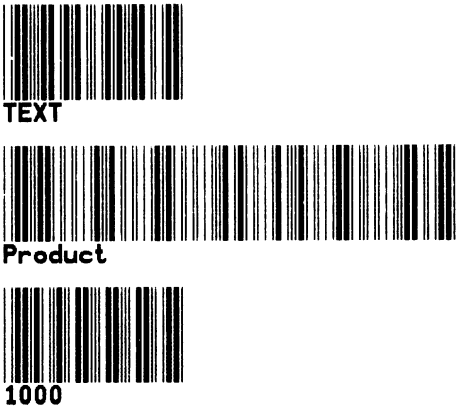
```
FK"TEST3" ↵
FS"TEST3" ↵
C0,6,L,+3,"Counter 0" ↵
B120,50,0,2,3,6,100,B,C0 ↵
B320,50,0,2,3,6,100,B,C0+1 ↵
B520,50,0,2,3,6,100,B,C0+2 ↵
FE ↵
```

Example 2:

This example illustrates how fixed text, variable text, and counters can be used in text fields in the Form Edit mode:

```
FK"TEST4" ↵
FS"TEST4" ↵
V00,25,l,"Product name" ↵
C0,4,L,+1,"Start serial No" ↵
B50,50,0,3,2,6,100,B,"TEXT" ↵
B50,200,0,3,2,6,100,B,V00 ↵
B50,350,0,3,2,6,100,B,C0 ↵
FE ↵
```

After retrieving and printing the form, the label may look like this:



Combination of several options can be used, for example:

```
B50,300,0,3,1,2,50,B,"Deluxe"V01C2"Combo" ↵
```

This example writes a Code 39 bar code containing the information "Deluxe", the contents of variable 01, the contents of counter 2, the text "Combo", and the contents of variable 01.

Bar Code 128 Family

The Code 128 family has three unique data character subsets: A, B and C. The choice of data character subset depends on the start character representing code A, B or C. The code subset can be redefined within the symbol by code set control characters A, B or C, or shift. In addition, a set of special functions are available via FNC1, FNC2, FNC3 and FNC4. All these can be invoked in the data field. The human readable part in the symbol, FNC and code subset control characters will be represented by a space. The input data string is a fixed string of maximum 64 alpha-numerical characters.

Code 128 With Auto Selection of Subset and UCC/EAN

Code 128 and EAN128 use automatic subset selection of Subset A, B, or C, both at the start and within the bar code.

Code 128 With Manual Selection of Subset

By starting the Code 128 symbol with manual selection of subset A, B or C, the user has complete control of composing the symbol data output. By invoking special control characters in the data string, it is possible to switch between subsets or other special cases inside the symbol, see table below.

String Invocation Rules

Command Type ▶	**Text Print**	**Bar Code Manual Selection**		**C128**	**Bar Code C128 Auto UCC/EAN128**
Command ▶	A	B	B	B	B, b[5]
Parameter p_4 ▶		1A	1B	1C	1,IE[4],1EC[5]
Starts with ▶		Start A	Start B	Start C	Start A, B, or C (+FNC1)
Invocation in String ▼	Text output ▼	Bar code output ▼	Bar code output ▼	Bar code output ▼	Bar code output ▼
\”	“	“	“	N/A[2]	“
\\	\	\	\	N/A[2]	\
\1	1	<FNC1>	<FNC1>	<FNC1>	<FNC1>
\2	2	<FNC2>	<FNC2>	NA[2]	<FNC2>
\3	3	<FNC3>	<FNC3>	NA[2]	<FNC3>
\4	4	<FNC4>	<FNC4	NA[2]	<FNC4>
\A	A	Ignore[1]	<Code A>	<Code A>	A
\B	B	<Code B>	Ignore[1]	<Code B>	B
\C	C	<Code C>	<Code C>	Ignore[1]	C
\S	S	<Shift>	<Shift>	N/A[2]	S
Char 06[3]	Char 06	Char 06	NA[2] in Code B	NA[2]	<FNC1>

Notes on the invocation rule table:

[1] Nothing happens if invocation code \A, \B, or \C refer to the same subset marked "Ignore".

[2] Not applicable combinations will give an error 01.

[3] Char hex 06 in string is a legacy for FNC1 invocation from earlier bar code 128 implementations.

[4] EAN/UCC symbology always included FNC1 in the first data position and do not support other FNCs.

[5] Linear component of 2D bar code EAN/UCC Composite-Code (Data 1).

b - Two-Dimensional Bar Codes, General Part

Purpose: This command is used to print complex bar codes such as Data Matrix, PDF417, MaxiCode, and QR Code. The command consists of a leading set of general position and bar type select parameters, and a trailing code specific part defining the bar code's appearance and its input data.

Syntax: bp_1, p_2 , p_3, [Code specific options]

Parameter Descriptions

Parameter	Description
p_1	Horizontal start position (x) in dots.
p_2	Vertical start position (y) in dots.
p_3	Code type: C = Selects EAN/UCC Composite-Code D = Selects Data Matrix M = Selects MaxiCode P = Selects PDF417 Q = Selects QR Code [Code specific options], see For help, see "b - EAN/UCC Composite-Code" on page 53.

Remarks: The standard program packages contain all the two-dimensional bar codes mentioned above.

If the amount of data will not fit in the area specified, the indicator will light orange, indicating an error.

b - EAN/UCC Composite-Code

Purpose: The following EAN/UCC Composite-Code bar code specific options should append the general part of the two-dimensional bar code command (indicated by the italicized characters, see **"b - Two-Dimensional Bar Codes, General Part" on page 52** for more information. Presently, only EAN/UCC-128 with CC-C is supported.

Syntax: bp_1,p_2, p_3, p_4,o,x,y,s,1,[B,][P][F,]M,"$Data_1$ | $Data_2$"

Parameter Descriptions

Parameter	Description
p_4	Selection of bar code type (linear part and two-dimensional part) 1EC and EAN/UCC-128 with CC-C
o	0 = No rotation 1 = 90° rotation clockwise 2 = 180° rotation clockwise 3 = 270° rotation clockwise Limit is 0-3. Default is 0.
x	Module width of pixel size in dots, magnification of both linear and 2D composite component. Limit is 1-10 Default is 2.
y	Bar height on linear part in dots. Limit is 2-999. Default is 100.

Parameter Descriptions (continued)

Parameter	Description
Y	Bar height on two-dimensional composite part in dots. Minimum height is 2*x, nominal is 4*x. Limit is 2-999. Default is 4 (2*x).
s	Separator pattern height in dots between linear component and two dimensional composite component. Limits is 1-60 (minimum s = 1*x, maximum s = 6*x). Default is same as x.
l	Manually controlled column count on CC-C code. If not set, the column count will be optimized to linear bar width. Limit is 1-30. Default is NA.
B	Flag for human readable text of linear component part, to present within the same rules as to the single bar code. Font selection depends on the bar code type. It is similar to B-command parameter p_8, see"B - Standard Bar Codes" on page 46. Default is no B flag.

Parameter Descriptions (continued)

Parameter	Description
px, y, m	Human readable text of two-dimensional composite component part appended by additional variables for any position relocation around the composite bar code: x = horizontal start location: max is 4 digits. y = vertical start location: max is 4 digits. m = maximum characters per line. The max is 3 digits, and 0 = no limit. Fixed Font selection 3 without any multiplier used. CR, LF in data will direct wrap around the rest of text to the next line. Similar design as b-command parameter p on PDF417. Limits are x and y position limitations. Default is no p.
F	Restrain control filter, to reduce input data information, by deleting special characters for the bar code rendering, but let them be a part of human readable text. Deleted characters are ( ) space CR and LF. Filters only hex 0ah, 0dh, 20h, 28h, 29h. Default is no F.
M	Manually controlled Pipe character: Separator between data of linear and two-dimensional composite part. Use this parameter only in exceptional case. Limit is any printable character. Default is M|.
$Data_1$|$Data_2$	The data area represents a string data field surrounded by quotation-marks. Max data length is relative chosen model type and data mode. Invocation code in string follows the rules for selectable Code 128, see"B - Standard Bar Codes" on page 46. FNC2, FNC3, and FNC4 are not allowed in composite code.

Parameter Descriptions (continued)

Parameter	Description
$Data_1$	String of primary data to linear component. Limit is 48 digit maximum.
|	Data separator pipe character, see
$Data_2$	string of secondary data to two-dimensional composite data. Limit is all characters are in ISO 646.

Errors will result if:

- data is not suitable to optional parameters => error 01 (syntax error).
- data is too long or out of range => error 03 (at length error).
- data does not fit in area specified => error 50.

Example:

```
N ↵
b10,20,C,1EC,x4,Y12,B,p40,210,0,F,"(01)938123456789
01|(10)ABCD123456(410)3898765432108" ↵
P
```

(01)93812345678901
(10)ABCD123456(410)3898765432108

b - Data Matrix

Purpose: The following Data Matrix specific options should append the general part of the two-dimensional code command indicated by the italicized characters, see **"b - Two-Dimensional Bar Codes, General Part" on page 52** for more information.

Syntax: $bp_1,p_2,p_3,[P_4,$"DATA"]

Parameter Descriptions

Parameter	Description
P_4	Narrow bar from 1-15
“DATA”	Represents a fixed data field, maximum 125 characters long

Example:

```
N ↵
b40,80,D,5,”12345678901234567890123456789012345678901234567890123456789
0
12345678901234567890”  ↵
P1 ↵
```

b - PDF417

Purpose: The following PDF417 bar code specific options should append the general part of the two-dimensional code command indicated by the italicized characters, see **“b - Two-Dimensional Bar Codes, General Part” on page 52** for more information.

Syntax: bp_1,p_2,p_3,[www,hhh,s,c,p,f,d,x,y,r,l,t,o],“DATA”

Parameter Descriptions

Parameter	Description
www	Maximum print width in dots (3 digits).
hhh	Maximum print height in dots (3 digits).
s	Sets error correction level. Legal values are 0-8. If level is not specified, a level that will generate about 1/8 as many ECC code words as data code words is selected.
c	Selects data compression method: 0 = Selects auto-encoding (default) 1 = Selects binary mode
p	Print human readable code appended by additional variables: xxx = Horizontal start location (3 digits). yyy = Vertical start location (3 digits). mmm = Maximum characters per line (3 digits).
f	Center pattern in area: 0 = The pattern will print upper left justified in the area defined by the w and h values. 1 = The pattern is printed in the middle of the area defined by the w and h values (default).
d-	Print code words: 0 = Values of code words not printed (default). 1 = Values of code words printed.
x-	Module width. Legal values are 2-9.
y-	Set bar height. Legal values are 4-99 dots high.
r-	Maximum row count, refer to PDF417 specifications.
l-	Maximum column count, refer to PD417 specifications.
t-	Truncated flag: 0 = Not truncated. 1 = Truncated.

Parameter Descriptions (continued)

Parameter	Description
o-	Rotation: 0 = 0° rotation clockwise. 1 = 90° rotation clockwise. 2 = 180° rotation clockwise. 3 = 270° rotation clockwise.
“DATA”	Represents a fixed data field.

Remarks: If parameter www (maximum print witch) gives less space than required by the sum of parameters x- (module witch) and l- (maximum column count), error condition 50 will occur.

Also, if parameter hhh (maximum print height) gives less space than required by the sum of parameters y- (set bar height) and r- (maximum row count), error condition 50 will occur.

Example:

```
N ↵
b40,40,P,400,300,p40,340,20,f1,x3,y10,r60,l5,“ABCDE
FGHIJK1234567890abcdefghijk” ↵
P1 ↵
```

ABCDEFGHIJK123456789
0abcdefghijk

Note: The last parameter in the b command (l5) is lowercase L and the digit 5, not 15.

b - QR Code

Description: QR Code is a matrix symbology defined in AIM International ITS/97-001.

QR Code Model 1 and Model 2 Descriptions

Characteristics	Model 1	Model 2
Model version	Original	Enhanced
Symbol size in cell (increment of 4 cells)	21*21 to 73*73	1‘*21 to 177*177
1. Numeric data modes: Digits 0-9	Maximum is 1167 characters.	Maximum is 7089 characters.
2. Alphanumeric data mode: Digits 0-9, upper case letters A-Z, and 9 other characters	Maximum is 707 characters.	Maximum is 4296 characters.
3. 8-bit byte data mode: JIS 8-bit character set (Latin and Kana) in accordance with JIS X 0201	Maximum is 486 characters.	Maximum is 2953 characters.
4. Kanji data mode: Shift JIS X 0208	Maximum is 299 characters.	Maximum is 1817 characters.
5. Mixing mode: Mix of mode 1-4 above	N/A	Applicable

Note: The maximum data length in ESIM is approximately 780 bytes.

Error Correction

Four levels of Reed-Solomon error correction allowing recover of:

- L = 7%
- M = 15%
- Q = 25%
- H = 30%

Masking (Intrinsic)

This enables the ratio of dark to light modules in the symbol to b approximated to 1:1 whilst minimizing the occurrence of arrangements of close modules which would prevent efficient decoding.

The QR bar code specific options should append the general part of the two-dimensional bar code command (indicated by the italicized characters, see **"b - Two-Dimensional Bar Codes, General Part" on page 52** for more information.

Syntax: bp_1, p_2, p_3,[cn,mn,on,sn,xn,ynn],"DATA"

Parameter Descriptions

Parameter	Description
c	Fixed data mode character set: 1 = Numerical data mode 2 = Alphanumerical data mode 3 = 8-bit data mode (JIS 0201) 4 = Kanji data mode (JIS 0208) 5 = Mixing mode (any of mode c1-c4, model type 2) Model 1 default = Most optimal alternative of c1 to c4. Model 2 default = c5. Internally during rendering the QR Code, the mode can be switched within a bar code to minimize the bit stream area. By using parameter cn, the data mode character set is fixed.
m	Model type is defined and unchanged within one QR Code. 1 = Original version 2 = Enhanced version (recommended). Default.

Parameter Descriptions (continued)

Parameter	Description
o	Rotation: 0 = 0° rotation clockwise. Default. 1 = 90° rotation clockwise. 2 = 180° rotation clockwise. 3 - 270° rotation clockwise.
s	Set correction level: L = 7% (default) M = 15% (default) Q = 25% (default) H = 30% (default)
x	Masking factor 0-7 or 8 (auto). Default is 8.
y	Cell pixel size magnification 02-64. Default is 10.
“DATA”	Represents a fixed data field. Maximum data length is relative chosen model type and data mode, but is limited to approximately 780 bytes. Ordinary string controls, like \\for\and\“for”.

Errors: If data is not suitable to optional parameters, Error 01 (syntax error will result).

If data is too long or out of range, Error 03 (data length error will result).

If data does not fit in area specified, Error 50 will result.

Example

```
N ↵
b40,40,Q,sM,x1,"ABCDEFGHIJKLMNOPQRSTUVWHYZ123456789
012345678901234ABCDEFGHIJKLMNOPQ
RSTUVWHYZ1234" ↵
P1 ↵
```

C - Counter

Purpose: This command is used to define one of maximum 10 automatic counters used in consecutive numbering applications, such as serial numbers. Counters can only be used in the Form Edit mode, and not in the Direct mode.

Syntax: $Cp_1,p_2,p_3,p_4,[p_5,],$"PROMPT"

Parameter Descriptions

Parameter	Description
p_1	Counter number (0-9).
p_2	Maximum number of digits for the counter (0-29).
p_3	Field justification: L = Left justification. R = Right justification. C = Center justification. N = No justification.
p_4	Step value. Plus or minus sign followed by a single digit (1-9): + Increase. - Decrease.

Parameter Descriptions (continued)

Parameter	Description
p5	Alphanumerical support (optional): N= Numerical support. Anything else than digit and leading spaces in DATA will give error 03. A = Alpha support: Base 10 on digits, base 26 on letters (default). Anything else than digit, leading spaces, and uppercase alpha characters in DATA will give error 03. Wrap around: 0 → 9 goes back to 0. A → Z goes back to A. B = Alphanumerical support: Base 36 Anything else than digit, leading spaces, and uppercase alpha characters in DATA will give error 03. Wrap around: 0 → 9A → Z. 9 goes to A. Z goes to 0.
[-]	A single leading minus sign in the prompt field will cause the prompt to be sent one time only after the form is retrieved (Keyboard display unit only, see below).
[--]	A double leading minus sign in the prompt field will cause the prompt to be suppressed (Keyboard display unit only, see below).
“PROMPT”	An ASCII text field that will be transmitted to the Keyboard display unit or host via the serial interface each time a form containing this command is retrieved. It usually requests the operator to enter the starting value for the counter.

Remarks: This command is used in forms that require sequential numbering. When initializing counters, they must be defined in order (for example C0, C1, C2, etc.) after possible variables.

To print the contents of the counter, the counter number (C0- C9) is entered in the "DATA" field of A (Print Text) or B (Print Bar Code) commands.

Prompts will only be displayed if a UI command has been issued after last power-up. The Keyboard display unit sends the UI command automatically.

The field justification parameter (p_3) affects the way the counter will be printed. When p_3 = L, R, or C, the counter value will be printed left, right or center justified in an area with a width defined by p_2 (number of digits). If no justification is selected (p_3 = N), the field will truncated from the right side so as to not exceed the set maximum field length, which may be useful when using a counter as input data to a bar code.

If the start value entered, when the form is retrieved for printing, is started by one or several zeros (0), the entire area specified by p_2 (number of digits) will be padded with leading zeros, that is p_3 (field justification) will have no effect.

Note: If a single counter is stepped up several times on the same form, then the step value p_4 must be set to the number of times the counter is used in the form or equivalent to what the step values for the single counter add up to in this form. A Cn±x command must also be used when designing the actual form. See the A and B commands.

The table below illustrates how leading space characters (“ ”) in DATA will be influenced by next character at wrap around. If next position is a digit with wrap around (9 → 0), space will be incremented to digit 1. If next position is a letter with wrap around (Z → A), space will be incremented to letter A.

P_5	Counter Command	Start Data	Data after p4 =+1		Data after p4 =+2	
N	C0,3,C,+1,N,“Numerical”	“99”	→	“100”	→	“101”
		“999”	→	“000”	→	“001”
A	C0,3,C,+1,A,“Alpha” (if no P5 parameter is included, method A is selected by default).	“ A9”	→	“ B0”	→	“ B1”
		“ Z9”	→	“AA0”	→	“AA1”
		“0Z9”	→	“1A0”	→	“1A1”
		“ ZZ”	→	“AAA”	→	“AAB”
		“ZZ9”	→	“AA0”	→	“AA1”
B	C0,3,C,+1,B,“Alpha-num”	“ 99”	→	“ 9A”	→	“ 9B”
		“ A9”	→	“ AA”	→	“ AB”
		“ 9Z”	→	“ A0”	→	“ A1”
		“ ZZ”	→	“100”	→	“101”
		“ZZZ”	→	“000”	→	“001”

Example:

```
FK“TEST5” ↵
FS“TEST5” ↵
C0,5,L,+1,“Start value CNT 0” ↵
C1,5,R,+1,N,“Start value N-CNT 1” ↵
C2,5,C,+1,A,“Start value A-CNT 2” ↵
C3,5,N,+1,B,“Start value B-CNT 3” ↵
A50,050,0,3,1,1,N,“Cnt Default, left →
→ justified :“C0”:” ↵
A50,100,0,3,1,1,N,"Cnt Numerical, right →
→ justified :“C1”:” ↵
A50,150,0,3,1,1,N,“Cnt Alpha, center →
→ justified :“C2”:” ↵
A50,200,0,3,1,1,N,"Cnt Alpha-num, not →
→ justified :“C3”:”
FE ↵
FR “TEST5” ↵
```

```
? ↵
A9 ↵
99 ↵
ZZ9 ↵
ZZ9 ↵
P3 ↵
```

Protect Counters

When the optional Keyboard display unit (KDU) is used, the label form can be designed to "skip" a consecutive number prompt, thereby protecting the data. This feature is especially useful when the counter represents a serial number or other types of number, that should never be repeated.

By placing one (1) minus sign as the first character of the prompt, the prompt will appear only once after the form is retrieved.

Example 1:

```
C0,10,L+1,"-Enter Serial Number:" ↵
```

By placing two (2) minus signs as the first two characters of the prompt, the prompt will never be displayed.

Example 2:

```
C0,10,L+1,"- -Enter Serial Number:" ↵
```

The protected consecutive number is accessed and modified from the optional Keyboard display unit only.

Enter the following when the Keyboard display is showing:

```
FORM - retrieve form
F2 - list forms vx.x
```

1. (Optional) Press the **Exit** key to display the above.
2. Press the **F1** key.
3. Press **4**, **9**, **1**, and **6**.
4. Press the **Form** key.
5. Key in the form name and press **Enter** to retrieve.
6. Enter or modify the consecutive number.
7. When complete, print label to store new number in memory.

C - Cut Immediate

Purpose: This command is used to make the cutter perform a cutting cycle without printing a form.

Syntax: C

Remarks: The command C is used for two purposes. C appended by parameters is used to define counters in the Form Edit mode, whereas C without any appending parameters is used to initiate a cutting operation.

C (Cut Immediate) cannot be used inside a form or in connection with a the Keyboard display unit.

To perform self-cleaning of the cutter blade, issue five consecutive C commands without any media loaded.

Example:

To perform a cutting cycle, enter the command below:

C ↵

D - Density

Purpose: This command is used to select the print density.

Syntax: Dp_1

Parameter Descriptions

Parameter	Description
p_1	Density settings (0-15). Default is 10. 0 is the lightest printing and 15 is the darkest.

Remarks: The density command is used to control the energy to the printhead. A number of factors affect the actual darkness of the printout:

- Direct thermal printing or thermal transfer printing.
- Print speed.
- Different brands of direct thermal media.
- Different combination between transfer ribbons and receiving face materials.

- Different ambient temperature and humidity.

The printed information may also require the density to be adjusted. Typically, this applies to different bar code orientations and densities. Refer to the tables in the pages that follow for recommended initial settings.

Test after the print speed has been set, see "S - Speed Select" on page 113, and make further adjustments until you have found the settings which best apply to your unique application.

Example:

To select a printhead density of 9, enter the command below:

`D9` ↵

Recommended density settings are identified in the following tables. Further adjustments might be necessary depending on print speed, bar code density, orientation, and ambient temperature/humidity conditions.

Recommended Density Settings For Direct Thermal Printing

Label/Tag Type	Recommended Density at Speed S = 2	Maximum Speed
Duratherm II	D10	S3
Duratherm II Tag	D9	S1
Duratherm Lightning	D9	S3
Duratherm IR	D7	S3
Thermal Top	D8	S3
Thermal Eco	D8	S3
Thermal Top Board	D11	S2
Thermal Eco Board	D7	S2
Thermal IR	D12	S3
Thermal Top High Speed	D6	S3
Duratherm Lightning	D9	S3

Recommended Density Settings For Thermal Transfer Printing

Label/Tag Type	Ribbon Type	Recommended Density at Speed S = 2	Maximum Speed
Duratran II	Standard	D4	S3
Duratran II Tag	Standard	D4	S2
Duratran II	Premium	D5	S3
Duratran II Tag	Premium	D6	S2
Kimdura	Premium	D6	S3
Matte Polyester	Premium	D6	S3
Gloss Polyester	Super premium	D7	S3
TTR Uncoated	GP02	D1	S2
TTR Matte Coated	HP05	D6	S3
TTR Premium	HP05	D4	S3
TTR Premium Board	HP05	D7	S1
TTR Polyethylene	HP05	D2	S3
TTR Gloss Polyethylene	HP05	D7	S3
TTR High Gloss White Premium	HP05	D7	S3
TTR Matte Coated	HP07	D7	S3
TTR Premium Board	HP07	D8	S1
TTR Polyethylene	HP07	D4	S3
TTR Gloss Polyethylene	HP97	D8	S3
TTR High Gloss White Premium	HP07	D9	S3
TTR High Gloss Polyester	HR03	D7	S3

EI - List Soft Fonts

Purpose: This command makes the printer print a list of all soft fonts that are stored in memory.

Syntax: FE

Remarks: This command is related to ES (Store Soft Fonts) and EK (Delete Soft Fonts).

Example:

To print a soft font list, enter the command below:

EI ↵

```
Soft font information:
z:011char,107dots,0dir
y:224char,039dots,0dir
x:224char,034dots,0dir
Soft font memory left:086K
```

EK - Delete Soft Font

Purpose: This command is used to delete soft fonts from memory.

Syntax: EK["name"|"*"]

Remarks: Soft fonts are stored using the ES command and listed using the EI command.

Soft fonts can also be deleted from the printer using a program such as Intermec LabelShop or Intermec InterDriver.

Example:

EK"a" ↵

EK"*" ↵

ES - Store Soft Font

Purpose: This command is used to download and store soft fonts in memory.

Syntax: ES"name"p_1p_2,$p_3a_1b_1c_1$"$data_{1...}a_nb_nc_n$"$data_n$"

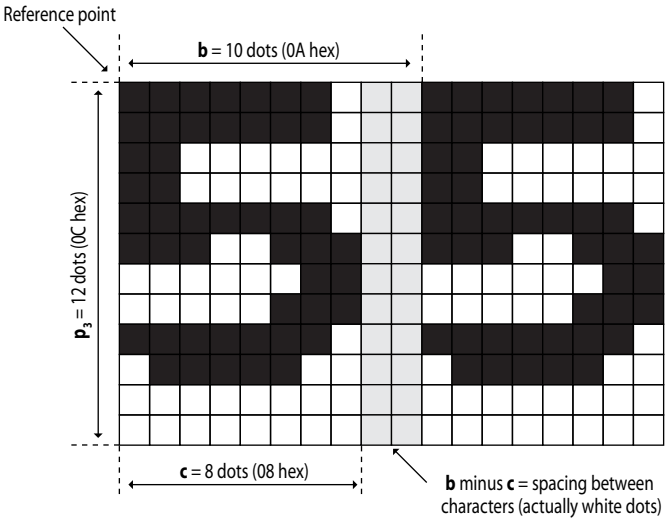
Parameter Descriptions

Parameter	Description
$name_{1-n}$	Name of the soft font (one lowercase letter only in the range a-z). Lower case named fonts minimize soft font memory usage to only store fonts downloaded and have 256 character limit.
p_1	Number of characters to be downloaded using hexadecimal coding. Range 00-FF hex (1-256 characters per soft font set).

Parameter Descriptions (continued)

Parameter	Description
p_2	Character rotation using hexadecimal coding: 00 hex =0° and 180° 01 hex = 90° and 270° clockwise 02 hex = All four directions (2 pairs)
p_3	Font height measured in dots and specified using hexadecimal coding. Range 00-FF hex. Font height includes accentors and dissenters of character and need to fit in the character cell of 256 dots = 32.03 mm (1.26 in).
a	Map position of character using hexadecimal coding. Range 00-FF hex.
b	Spacing to next print character in dots using hexadecimal coding.Range 00-FF hex. Must be greater than or equal to the character width specified by parameter c.
c	Width of character in dots using hexadecimal coding. Range 00-FF hex.
"data"	p3 x c1 = bit map data (in bytes). Data is received in bytes on a line by line basis. The font character's 0,0 cell map position is in the top left corner of the map as viewed in the 0 degree rotation. Repeat parameters a, b, c, and data for each character until all characters in the set have been downloaded. For fonts with the rotation parameter p_2 set to 02 hex (all directions), repeat the individual font character download for each 90° rotated character from the start of the character set until all rotated characters in the set have been downloaded. The number of individual character maps downloaded will be double the characters in the font set (p_1).

Remarks:

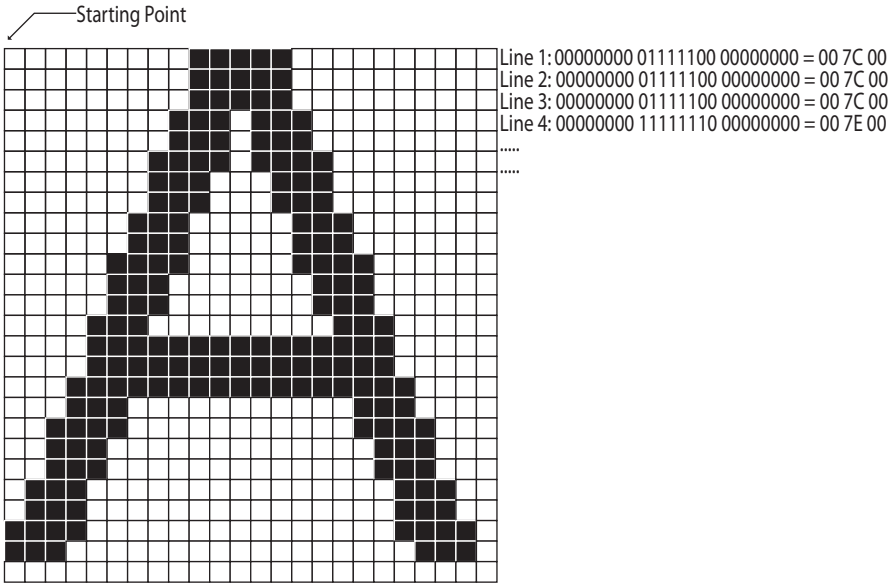


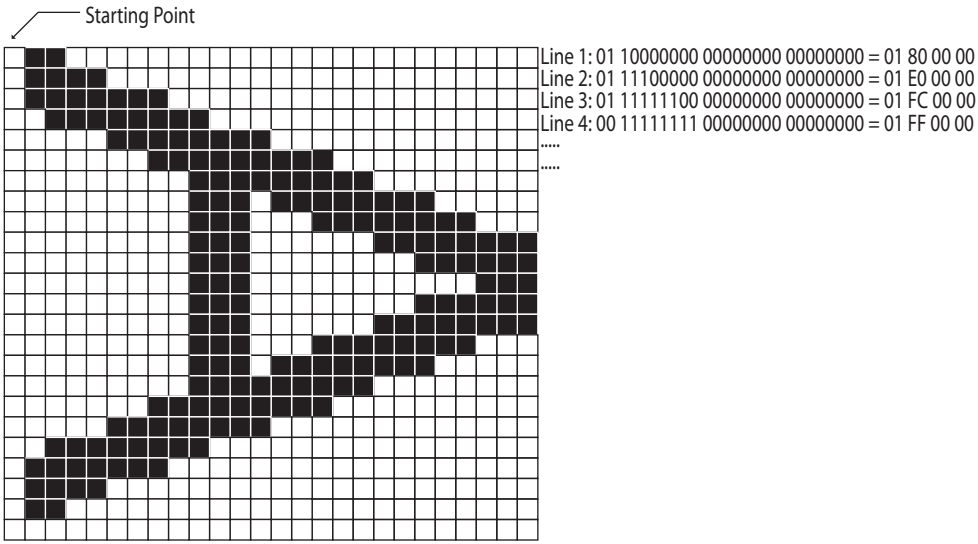
This picture illustrates the parameter p_3, b, and c.

The black and white bitmap that represents the character must be converted to hexadecimal code. The bitmap is described line by line from left to right, starting from the upper left corner of the character cell. A white dot is represented by 0 and a black dot by 1. Each byte (that is 8 dots) will thus form a binary number, that is converted to hexadecimal code. The last byte in a line is padded with zeros to complete the line and data byte. The data is the sent to the printer as a continuous string of hexadecimal byte representations in line order.

Soft fonts can also be downloaded to the printer using Intermec Labelshop or Intermec Interdriver.

Soft fonts can be listed using the EI command.





These illustrations show how a character in 0° and 90° rotation is downloaded to the printer.

eR - User-Defined Error/Status Character Control

Purpose: This command controls the format of error messages, mostly when command US has enabled the printer's status reporting feature, and also at some high priority error types when command UN has disabled status report.

Syntax: $eRp_1,p_2,[,p_3]$

Parameter Descriptions

Parameter	Description
p_1	Reply error action single character when error occurs: Value of p_1 is one single ASCII character, except hex 00. Default is 0.
p_2	Mode of User Defined Error/Status Character reply (one ASCII numerical: p_2 = 0 Mode of Reply Default, where error number is NN: Output at error action: <NAK>NN<XOFF> Output at error recovery: <XON> p_2 = 1 Mode of Reply only with character, defined by p_1: Output at error action: <p_1><CRLF> Output at error recovery: <p_3><CRLF> p_2 = 2 Mode of Reply with character p_1 and error number NN: Output at error action: <p_1>NN<CRLF> Output at error recovery: <p_3>00<CRLF> p_2 = Mode of <XOFF> and <XON> replacement: Like default mode (0), but with all <XOFF> replaced by p_1 and all <XON> replaced by optional p_3: Output at error action: <NAK>NN<p_1> Output at error recovery: <p_3>NN
p_3	Optional recovery action single character: The normal recovery character <XON> will be replaced by p_3. Value of p_3 is one single ASCII character, except hex 00. If no p_3 is available, p_1 is used instead as a recovery action character.

Remarks: Exception of output: Error 07 "Out of media or ribbon," is added after 07 with Pnnn or Rnnn, where nnn = number of remaining labels:

Output at error media action (mode 0):

`<NAK>07Pnnn<XOFF>`

Output at error ribbon action (mode 0):

`<NAK>07Rnnn<XOFF>`

Output at error recovery:

`<XON>`

Mode 1, 2, and 6 changes the output in a similar way.

All parameters are saved in Flash memory. If more than one error occurs at the same time, there will be an output for each error recovery, like `<p1>00<CRLF>` on mode 2. Refer to Appendix A for more information on error handling.

Example:

Command	Manual Action	Reply	Comments
US ↵		<ACK>	Enable error reporting.
eRC,2,Q ↵			Reply error with character "C" and error NN and at recovery with character "Q".
	Open cover	C11<CRLF>	Manual action generates error 11.
	Close cover	Q00<CRLF>	Manual recovery on error 11.

f - Cut Position

Purpose: This command is used to adjust the cutting position along the y-axis in relation to the media according to other characteristics of individual printers and batches of media.

Syntax: fp_1

Parameter Descriptions

Parameter	Description
p_1	Cut position index measured in dots: **203.2 dpi (8 dots/mm):** Recommended: 070 - 130 Minimum/Maximum: 020 - 180 Default is 100 **300 dpi (11.81 dots/mm):** Recommended: 050- 150 Minimum/Maximum: 000 - 200 Default is 100

Remarks: When using labels on liner, the printer will advance each printed label to the cutting position (between two labels) according to the program before the liner is cut. Due to differences between batches of media, the printer may not cut exactly between labels, but either cut a slice at the last part of the printed label or a slice at the front part of next label.

Cutting through labels should be avoided, because the adhesive will stick to the cutting parts and prevent them from operating properly.

If the cut occurs in the already printed label, increase the cut position index value.

If the cut occurs in the next label, decrease the cut index value.

Example:

To increase the cut position index value, enter the command below:

`f110` ↵

FE - End Store Command

Purpose: This command is used to end a Form Store sequence.

Syntax: FE

Remarks: The Form Store sequence must start with the FS command and end with the FE command.

Example:

```
FS"formname" ↵
........
FE ↵
```

FI - Print Form Information

Purpose: This command makes the printer produce a list of all forms stored in memory.

Syntax: FI

Remarks: The FI command will be executed directly, without appending any Linefeed.

Issue a FI command after having stored a form to make sure the storing was successful and to check the amount of free form memory.

FK - Delete Form

Purpose: This command is used to delete a specified form or all forms from memory.

Syntax: FK"name"|"*"

Parameter Descriptions

Parameter	Description
"name"	By entering a name of a form, that form will be deleted from memory.
"*"	By entering an asterisk (*) as a wildcard, all forms will be deleted from memory.

Example 1:

To delete a form named FORM1, enter the command below:

```
FK"FORM1" ↵
```

Example 2:

To delete all forms, enter the command below:

```
FL"*" ↵
```

FR - Retrieve Form

Purpose: This command is used to retrieve a form that was previously stored in memory.

Syntax: FR“name”

Parameter Descriptions

Parameter	Description
“name”	This is the form name used when the form was stored. The printer is case sensitive.

Remarks: To print a list of forms currently stored in memory, use the FI command.

Example:

To retrieve the form named “Test1,” enter the command below:

```
FR"Test1" ↵
```

FS - Form Store

Purpose: This command is used to begin a Form Store sequence.

Syntax: FS“name”

Parameter Descriptions

Parameter	Description
“name”	This is the form name used when retrieving the stored form. The name may be from 1-8 characters. The printer is case sensitive (the form names will be stored with the exact case entered on the FS command line).

Remarks: All commands following FS will be stored in the Forms memory until a FE command is received, ending the form store process. If a form with the same name is already stored in memory, the FS command will result in an error and the old form will be retained. When updating a form, use the FK command to delete the old version before storing the new version. To print a list of the forms currently stored in memory, use the FI command.

Global commands, such as EI, EK, ES, FI, FK, GI, GK, GM, I, N, P, U, UE, UF, UG, UI, UM,UP, UN, US, Y, W, ?, or ^@ should not be used in a form store sequence.

Example:

This example begins the form store sequence of TEST1 and the ends it.

```
FS"TEST1" ↵
.....
FE ↵
```

Startup Form

A special case of forms is the startup form, that is automatically retrieved and prompted for variables (if necessary) each time power is applied to the printer. A startup form is created by naming the form "AUTOFR". To exit the "AUTOFR" mode, send XOFF or NULL to the printer on the serial interface.

Always test the form using another name before making it a startup form. If a startup form causes an error, there are two ways of clearing it:

- If the indicator lamp shines green, send XOFF or NULL to exit “AUTOFR” mode. Then delete the startup file using FK“AUTOFR”.
- If the indicator lamp shines orange, there is no communication and the memory must be erased by pressing the **Feed** key for more than 3 seconds in Dump mode.

GG - Print Graphics

Purpose: This command is used to print a graphic that has been previously stored in memory.

Syntax: GGp_1,p_2,“name”

Parameter Descriptions

Parameter	Description
p_1	Horizontal start position (X) in dots.
p_2	Vertical start position (Y) in dots.

Parameter Descriptions (continued)

Parameter	Description
"name"	This is the name used when the graphic was stored. The name may be from 1 to 8 characters. The printer is case sensitive, so the use of upper and lowercase letters must match the original name.

Remarks: A graphic can only be printed in same direction and size as when it was saved. There are no means of magnification or rotation of an individual graphic. However, the entire print image including all text, bar codes, graphics, lines, and boxes can be rotated 180° using the Z command.

Example:

This example prints the graphic "LOGO," see command below:

```
GG50,50,"LOGO" ↵
```

GI - Print Graphics Information

Purpose: This command will cause the printer to print a list of all graphics stored in memory.

Syntax: GI

Remarks: The GI command will be executed directly, without appending any Linefeed.

Issue a GI command after having stored a graphic to make sure the storing was successful, and to check the amount of free graphic memory.

GK - Delete Graphics

Purpose: This command is used to delete a specified graphic or all graphics from memory.

Syntax: GK"name"|"*"

Parameter Descriptions

Parameter	Description
“name”	By entering a name of a form, that form only will be deleted from memory.
“*”	By entering an asterik (*) as a wildcard, all forms will be deleted from memory.

Example1:

To delete the graphic named “LOGO,” see the command below:

GK“LOGO” ↵

Example 2:

To delete all graphics, see the command below:

GK“*” ↵

GM - Store Graphics In Memory

Purpose: This command is used to store PCX graphic files in Flash memory.

Syntax: GM“name”p_1“DATA”

Parameter Descriptions

Parameter	Description
“name”	This is the name that will be used when retrieving the stored graphic (maximum of 8 characters). The printer is case sensitive, that is, graphic names will be stored with the exact case entered on the GM command line.
p_1	This is the size of the original .PCX file in bytes. In DOS, the DIR command can be used to determine the exact file size.
“DATA”	The graphics data in 1-bit (black and white) PCX format. The resolution of the graphics must match resolution of the printer.

Remarks: The GM command saves the graphics in the Flash memory, so it will not be lost at power off. Use it for graphics that are used frequently and do not change, for example the logotype of your company. Compare with GW command. In a DOS system, the "DATA" portion can be sent to the printer via the parallel port using the DOS COPY command.

Example:

Assume you have a PCX file named LOGO.PCX in your current directory. Use a text editor to create a text file called for example STOREIT.TXT and store it in the same directory as the .PCX file.

```
↵

GM"LOGO"1421 ↵
```

To store the image in the default printer, at the the DOS prompt type:

```
COPY STOREIT.TXT PRN
COPY LOGO.PCX PRN .b
```

To store the image in the printer connected to port LPT1, at the DOS prompt type:

```
COPY STOREIT.TXT LPT1:
COPY LOGO.PCX LPT1: /b
```

The GI command can be used to verify that the graphic was successfully stored. If not, check that the .PCX file is in 1-bit (black and white) format and that the printer's free graphics memory is large enough to accommodate the graphics.

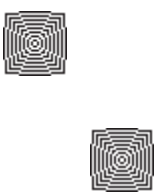
The following sequences show how to generate the image "PYRAM" and print two copies. Image "PYRAM" is saved as a PCX-file (version 2) in monochrome mode with a size of 32*32 dots. Data below is shown in hex and ASCII format:

0000	0D 0A 47 4D 22 50 59 52 41 4D 22 32 39 34 0D 0A	..GM"PYRAM"294..	Save image
0010	0A 02 01 01 00 00 00 00 1F 00 1F 00 0E 00 0E 00	.....	PCX-header
0020	00 00 00 FF FF FF 00 00 1D 04 00 00 00 00 00 00	.....	
0030	38 42 13 00 EC E9 12 00 03 00 00 00 90 41 13 00	8B.....A..	
0040	FF FF FF FF 00 00 00 00 FF FF FF FF DC E9 12 00	.....	
0050	00 01 04 00 01 00 00 00 00 00 00 00 00 00 00 00	.....	
0060	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	.....	
0070	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	.....	
0080	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	.....	PCX-header
0090	C1 FF C1 00 00 C1 FF 80 C1 FF C1 FF 01 BF 80 01	.....	end
00A0	C1 FD A0 7F C1 FE 05 AF C1 C0 03 C1 F5 A8 3F C1	.....?.	PCX data
00B0	FC 15 AB C1 E0 07 C1 D5 AA 1F C1 F8 55 6A C1 F0	.....Uj..	
00C0	0F 56 5A 8F C1 F1 5A 56 B8 1D 6A 55 A7 C1 E5 AA	.VZ...ZV..jU....	
00D0	55 6C 36 AA 55 5B C1 DA AA 55 56 6A AA C1 55 55	Ul6.U[...UVj..UU	
00E0	C1 AA AA C1 55 55 C1 AA AA 55 56 6A AA 55 5B C1	....UU...UVj.U[.	
00F0	DA AA 55 6C 36 AA 55 A7 C1 E5 AA 56 B8 1D 6A 5A	..Ul6.U....V..jZ	
0100	8F C1 F1 5A 6A C1 F0 0F 56 AA 1F C1 F8 55 AB C1	...Zj...V....U..	
0110	E0 07 C1 D5 A8 3F C1 FC 15 AF C1 C0 03 C1 F5 A0	.....?.....	
0120	7F C1 FE 05 BF 80 01 C1 FD 80 C1 FF C1 FF 01 C1	.....	
0130	FF C1 00 00 C1 FF	.....	
			PCX data end

Now the image is stored on FLASH and can be used anywhere and several times on the printout, as this example below:

```
N ↵
GG0,0,"PYRAM" ↵
GG50,50,"PYRAM" ↵
P ↵
```

The result will be a printout shown below:



GRP - Reply Graphics From Print Buffer

Purpose: This command is used to dump part of the printer buffer back to the host via the serial port.

Syntax: GRPp_1,p_2,p_3,p_4[,p_5]

Parameter Descriptions

Parameter	Description
p_1	X-position in printer dots
p_2	Y-position in printer dots.
p_3	Number of bytes across the graphic (8 dots = 1 byte).
p_4	Number of dot rows going down the graphic.
p_5	Parameter setting the format of the data. Allowed values are: B = Binary, not inversed. b = Binary, inversed (default). H = Hexadecimal, not inversed. h = Hexadecimal, inversed.

Remarks: This command is the reverse of the GW command and can be used to verify that data has been rendered correctly. What part of the print buffer is sent back to the host is specified by the the parameters p_1-p_4 above, and the format of this data by p_5.

Note: There is no flow control or handshaking when sending this data to the host, so use this command only for small areas.

Example:

`GRP20,10,4,32,h` ↵

The command above will give a reply in hexadecimal format like the one shown below.

```
00 FF FF 00 3F FF FF FE 5F FF FF FE 6F FF FF FE
77 FF FF FE 7B FF FF FE 7D FF FF FE 7E FF FF FE
FF 7F FF FF FF BF FF FF FF DF FF FF FF EF FF FF
FF F7 FF FF FF FB FF FF FF FD FF FF FF FE FF FF
FF FF 7F FF FF FF BF FF FF FF DF FF FF FF EF FF
FF FF F7 FF FF FF FB FF FF FF FD FF 7F FF FE FE
7F FF FF 7E 7F FF FF BE 7F FF FF DE 7F FF FF EE
7F FF FF F6 7F FF FF FA 7F FF FF FC 00 00 00 00
```

GW - Store Graphics In Image Buffer

Purpose: This command is used to store binary graphics files directly in the image buffer.

Syntax: GW p_1,p_2,p_3,p_4,"DATA"

Parameter Descriptions

Parameter	Description
p_1	X-position in printer dots
p_2	Y-position in printer dots.
p_3	Number of bytes across the graphic (8 dots = 1 byte).
p_4	Number of dot rows going down the graphic.
"DATA"	The graphic data in 1-bit (black and white) binary format. First data-byte represent dots in upper left corner and next data-byte next to the right. "1" bits are white dot-positions and "0"-bits are black dot-positions. The total amount of "DATA" bytes are p_3 * p_4.

Remarks: Use this command instead of GM for temporarily used graphics, for example images that change between each label. Not only is this method faster, but it also prolongs the life of the flash memory as the graphics are downloaded directly to SRAM.

The printer's firmware will calculate exactly how much data to expect based on p_3 and p_4.

Example:

The following sequence shows how to generate and print an image of a reduced square of 32*32 dots, with a diagonal line, all lines 1 dot wide. Data below is shown in hex format:

```
0000  4E 0D 0A 47 57 32 30 2C 31 30 2C 34 2C 33 32 2C  N..GW20,10,4,32,
0010  00 FF FF 00 3F FF FF FE 5F FF FF FE 6F FF FF FE  ....?...._...o...
0020  77 FF FF FE 7B FF FF FE 7D FF FF FE 7E FF FF FE  w...{...}...~...
0030  FF 7F FF FF FF BF FF FF FF DF FF FF FF EF FF FF  ................
0040  FF F7 FF FF FF FB FF FF FF FD FF FF FF FE FF FF  ................
0050  FF FF 7F FF FF FF BF FF FF FF DF FF FF FF EF FF  ................
0060  FF FF F7 FF FF FF FB FF FF FF FD FF 7F FF FE FE  ................
0070  7F FF FF 7E 7F FF FF BE 7F FF FF DE 7F FF FF EE  ...~............
0080  7F FF FF F6 7F FF FF FA 7F FF FF FC 00 00 00 00  ................
0090  50 0D 0A                                         P..
```

The result will be a printout shown below:

I - Character Selection

Purpose: This command is used to select the proper character set.

Syntax: Ip_1,p_2,p_3

Parameter Descriptions

Parameter	Description
p_1	Number of data bits (7 or 8). Default is 8.
p_2	Printer code page (1 digit, see table on next page). Default is 0.
p_3	KDU country code (3 digits, see table on next page. Default is 001 (only if p_1 = 8).

Printer Code Page (p_2) for 7 Data Bits ($p_1 = 7$)

P_2	Language
0	English (US)
1	English (UK)
2	German
3	French
4	Danish
7	Swedish
8	Swiss

Printer Code Page (p_2) for 8 Data Bits ($p_1 = 8$)

P_2	Code Page	Language
0	DOS 417	English (UK)
1	DOS 850	Multilingual (Latin 1)
2	DOS 852	Latin 2 (Cyrillic II/Slavic)
3	DOS 860	Portuguese
4	DOS 863	Canadian (French)
5	DOS 865	Norwegian
6	DOS 857	Turkish
7	DOS 861	Icelandic
8	DOS 862	Hebrew
9	DOS 855	Cyrillic
10	DOS 866	Cyrillic CIS 1
11	DOS 737	Greek
12	DOS 851	Greek 1
13	DOS 869	Greek 2
A	Windows 1252	Latin 1
B	Windows 1250	Latin 2
C	Windows 1251	Cyrillic
D	Windows 1253	Greek
E	Windows 1254	Turkish
F	Windows 1255	Hebrew

Note: In case code pages DOS 437, DOS863, or DOS 865 cannot produce the desired characters, use code page DOS 850 Multilingual.

KDU Country Code (p_3)

Code	Country
001	U.S.A
002	Canada
003	Latin America
027	South Africa
031	Netherlands
032	Belgium
033	France
034	Spain
039	Italy
041	Switzerland
044	United Kingdom
045	Denmark
046	Sweden
047	Norway
049	Germany
351	Portugal
358	Finland

For additional code page examples, refer to Chapter 9 on page 141.

Example:

To select an 8 bit character set for use in Sweden with a Keyboard display unit, see the command below:

```
I8,1,0486 ↵
```

i - Asian Character Spacing

Purpose: This command places an adjustable inter-character space between Asian font characters. The inter-character spacing gets multiplied with the text string by the selected font's horizontal and vertical multiplier values. For help, see **"A - Printing Text" on page 41.**

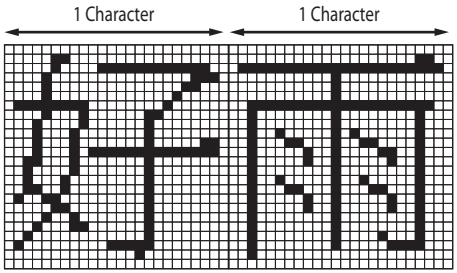
Syntax: ip_1

Example:

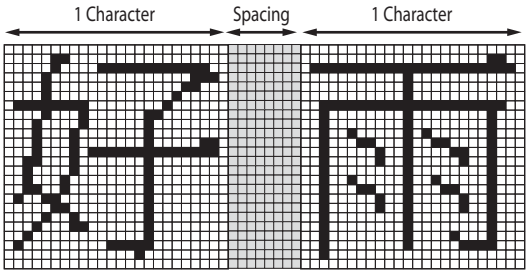
To select 8 dot spacing between Asian characters, see command below:

```
i8 ↵
```

i Command
Parameter p_1 set to 0 (default)



i Command
Parameter p_1 set to 8 (dots)



JB - Disable Top of Form Backup

Purpose: This command disables automatic top of form backup of the media.

Syntax: JB

Remarks: Top of form backup is used in connection with the j command, which makes the printer feed out an extra amount of media after printing the label, so as to allow the media to be torn or peeled off properly. It does not work with a cutter.

By default, the media is pulled back before printing the first label in next batch as to allow the printing to start at the top of the label, see JF command.

The JB command will disable this function, that is any j command will be ignored, and the printer will stop feeding when the end of the label becomes aligned with the printhead's dot line. However, the j command is kept stored in memory and can be enabled again using a JF command. To disabled top of form backup with a cutter, first send an O command followed by a JB command.

JF - Enable Top of Form Backup

Purpose: This command enables automatic top of form backup of the media.

Syntax: JF

Remarks: Top of form backup is used in connection with the **j** command, which makes the printer feed out an extra amount of media after printing the label, as to allow the media to be torn, peeled, or cut off properly.

By default, top of form is enabled, that is the media is pulled back before printing the first label in next batch as to allow the printing to start at the top of the label.

Top of form backup can be disabled by a JB command, that is, any j command will be ignored, and the printer will stop feeding when the end of the label becomes aligned with the printhead's dot line. In case of a cutter, the JB command must be preceded by an O command. However, the j command is kept stored in memory and can be enabled again using a JF command.

Activating the cutter using an OC command has the same effect as issuing a JF command.

j - Media Feed Adjustment

Purpose: This command makes it possible to set the media feed for tear-off (straight-through), peel-off (self-strip), or cut-off operation.

Syntax: jp_1

Parameter Descriptions

Parameter	Description
p_1	Length of media feed after printing in dots (0-160). Recommended values at 203.2 dpi (8 dots/mm): • Tear-off (straight-through) operation: 136 (default) • Peel-off (self-strip) operation: 110 Recommended values at 300 dpi (11.81 dots/mm): • Tear-off (straight-through) operation: 204 (default) • Peel-off (self-strip) operation: 165

Remarks: When using peel-off operation, the labels should remain slightly stuck to the liner (backing paper) so they do not fall off by their own weight, still can be manually removed with ease.

In case of tear-off operation, the media should be fed so the perforation between tags or the gap between labels become aligned with the tear bar. The j command allows the media feed to be adjusted accordingly, that is after the printer has been printed and the rear edge becomes aligned with the printhead's dot line, an extra amount of media feed is performed.

Caution

Do not use extremely small or large values for the j command, since they may cause the printer to feed or pull back the media continuously.

The extra media feed set by the j command can be enabled or disabled using JF and JB "Top of Form Backup" commands. By default "Top of Form Backup" is enabled.

Example 1:

This example shows the adjustment for peel-off operation at 8 dots/mm, see the command below:

```
j110 ↵
```

Example 2:

This example shows the adjustment for peel-off operation at 8 dots/mm, see the command below:

```
j136 ↵
```

LE - Line Draw Exclusive

Purpose: This command is used to draw black lines where the line will be white when intersecting a black area, or object, and vice-versa.

Syntax: LEp_1,p_2,p_3,p_4

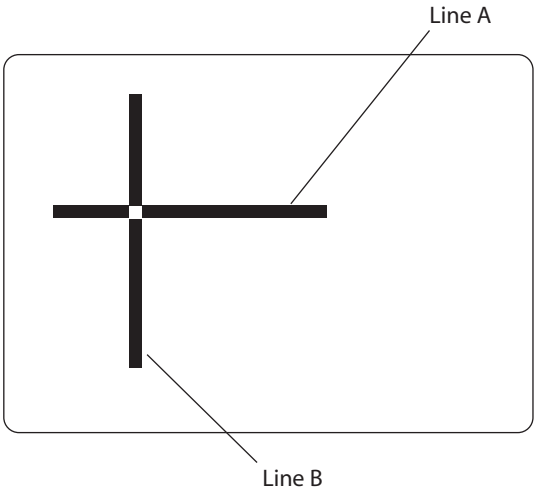
Parameter Descriptions

Parameter	Description
p_1	Horizontal start position (X) in dots.
p_2	Vertical start position (Y) in dots.
p_3	Horizontal length in dots
p_4	Vertical length in dots

Example:

In this example, we will draw two lines (A and B) intersecting one another, see the commands below:

```
N ↵
LE50,200,400,20 ↵
LE200,50,20,400 ↵
P1 ↵
```



LO - Line Draw Black

Purpose: This command is used to draw black lines, overwriting previous information.

Syntax: LOp_1,p_2,p_3,p_4

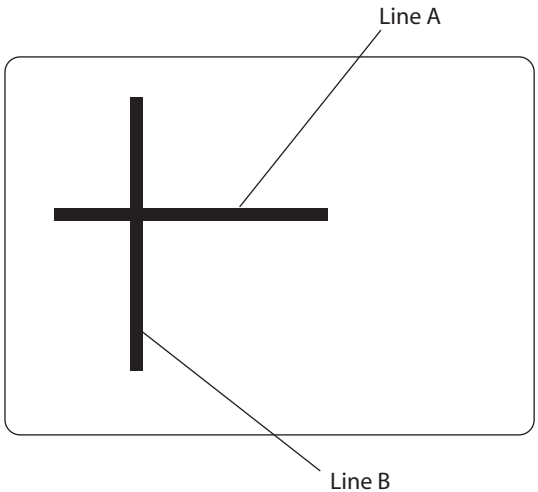
Parameter Descriptions

Parameter	Description
p_1	Horizontal start position (X) in dots.
p_2	Vertical start position (Y) in dots.
p_3	Horizontal length in dots
p_4	Vertical length in dots

Example:

In this example, we will draw two lines (A and B) intersecting one another, see the commands below:

```
N ↵
LE50,200,400,20 ↵
LE200,50,20,400 ↵
P1 ↵
```



LS - Line Draw Diagonal

Purpose: This command is used to draw diagonal black lines overwriting previous information.

Syntax: LSp_1,p_2,p_3,p_4,p_5

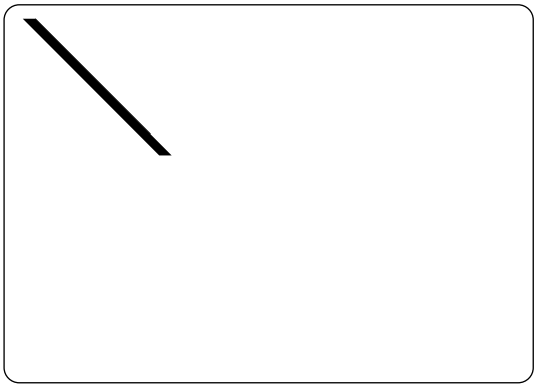
Parameter Descriptions

Parameter	Description
p_1	Horizontal start position (X) in dots.
p_2	Vertical start position (Y) in dots.
p_3	Line thickness in dots.
p_4	Horizontal end position (X) in dots.
p_5	Vertical end position (Y) in dots.

Example:

This examples shows you how to draw a diagonal line, see commands below:

```
N ↵
LS10,10,20,200,200 ↵
P1 ↵
```



LW - Line Draw White

Purpose: This command is used to draw white lines, effectively erasing previous information.

Syntax: LWp_1,p_2,p_3,p_4

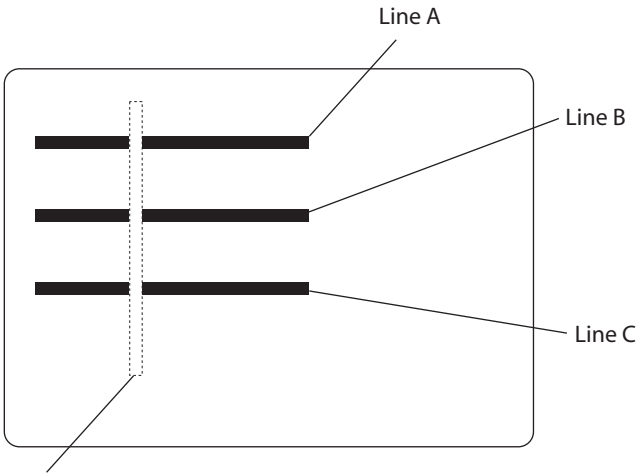
Parameter Descriptions

Parameter	Description
p_1	Horizontal start position (X) in dots.
p_2	Vertical start position (Y) in dots.
p_3	Horizontal length in dots.
p_4	Vertical length in dots.

Example:

This examples draws one white line (D) intersecting three black lines (A-C), see commands below:

```
N ↵
LO50,100,400,20 ↵
LO50,200,400,20 ↵
LO50,300,400,20 ↵
LW200,50,20,400 ↵
P1 ↵
```



Line D (the dotted lines will not be printed).

Memory - Memory Allocation

Purpose: This command is used to select the memory allocation used by the printer for data storage.

Note: This command only works with the PF8t or a printer that has ESim version 7.00 or greater with a CF card option installed.

Syntax: Mp_1

Parameter Descriptions

Parameter	Description
p1	Memory select value: 0 = Select printer's memory (default). 1 = Select the CF cards memory. F = Format the CF card.

Example:

```
MO ↵
UM ↵
UM150000,2048,0,0,0,0,0,0,UI80,001 ↵
```

Now, insert a 256 MB CF card into the printer and type:

```
UM150,249912,0,0,0,0,0,0,0,UI80,001 ↵
```

N - Clear Image Buffer

Purpose: This command is used to clear the image buffer before building a new image. It also clears any error list that has not been presented.

Syntax: N

Remarks: The N command is essential when printing labels in Direct mode. It is not necessary to use the N command before printing a form. The N command must not be used inside a form in the Form Edit mode.

O - Option Select

Purpose: This command is used to enable or disable various sensors and the cutter.

Syntax: O[Sp,N[,D[,Cnnn|Cb]]]

Parameter Descriptions

Parameter	Description
S	Enable reverse gap sensing.
N	Disable label taken sensor.

Parameter Descriptions (continued)

Parameter	Description
D	Disable ribbon end sensor.
Cnnn	Print nnn labels (1-255) before cutting.
Cb	Print batch before cutting

Remarks: Parameters can be entered in any order. Default is on.

S - Reverse Gap Sensing Enabled

This parameter reverses the operation of the label gap sensor so it interprets a blockage of light as a gap between labels or similar. Before using the S parameter, make sure to load the printer with the appropriate type of media. By default, the sensor will interpret blockage of light as a label or similar.

N - Label Taken Sensor Disable

When the label taken sensor is enabled, the communication to the printer will be busy as long as the sensor detects a label in the outfeed slot (does not work with a cutter, use ON to disable the sensor).

D - Ribbon End Sensor Disable

The ribbon end sensor (thermal transfer models only) detects reflections from the trailing silvery part of the transfer ribbon.

Direct Thermal Mode

If the ribbon end sensor is disabled parameter, density settings are interpreted via a table that gives an optimized printout for direct thermal printing. For help with density settings, see **"D - Density" on page 68**.

Thermal Transfer Mode

If the ribbon end sensor is enabled, an error will occur if no reflection is detected. The density settings are interpreted via a table that gives an optimized printout for thermal transfer printing. For help with density settings, see **"D - Density" on page 68**.

C - Cutter Enabled

If option C is enabled, cutting will be performed after print. Commands JF and JB will be disabled, but their values will remain stored in memory.

If option C is disabled, no cutting will be performed after print. Commands JF and JB will be enabled. Command f will be disabled but its value will remain stored in memory.

Combination of Various O Command Parameters

Command	LTS (N)	DT/TTR Mode (D)	Reverse Gap (S)	Cutter (C)
O	Enable	TT mode	Disable	Disable
OD	Enable	DT mode	Disable	Disable
OS	Enable	TT mode	Enable	Disable
OD,S	Enable	DT mode	Enable	Disable
OS,Cnnn|b	Enable	TT mode	Enable	Enable
OD,S,Cnnn|b	Enable	DT mode	Enable	Enable
ON	Disable	TT mode	Disable	Disable
ON,D	Disable	DT mode	Disable	Disable
ON,D,S	Disable	DT mode	Enable	Disable
ON,cnnn|b	Disable	TT mode	Disable	Enable
OCnnn|b	Not used	TT mode	Disable	Enable
OD,Cnnn|b	Disabled	DT mode	Disable	Enable

Example:

This example shows you how to enable DT mode, disable LTS, reverse gap sensor, and the cutter. See the command below:

`ON,D` ↵

oR - Character Substitution

Purpose: This command allows the advanced programmer to substitute the Euro currency character (€) for any ASCII character in printer-resident fonts 1-5. The original character can be restored by sending the oR command.

Syntax: oR[p_1[,p_2]]

Parameter Descriptions

Parameter	Description
p_1	If p_1 = E, the Eurocharacter will be mapped to the code page position specified by p_2. If no p_1 or p_2 parameters are give, all code pages will be reset to the original default character mapping.
p_2	Specifies the code page position for the Euro character in the range ASCII 32-255 decimal for all code pages, provided p_1 = E. If p_2 is omitted, the Euro character will be mapped to the code page position ASCII 213 decimal for all code pages provided that p_1 = E.

Remarks: The oR command is a global printer command and:

- It cannot be issued inside a form.
- It must be issued prior to issuing a text command and printing it.
- It affects a single character on all code pages. Changing the character position will restore the original character.
- Flash memory printer parameter data are preserved until they are changed by the oR command, or the printer is reset to default.

Example:

```
oRE ↵
oRE,128 ↵
oR ↵
```

P - Print

Purpose: This command is used to print the contents of the image buffer.

Syntax: Pp1[,p2]

Remarks: The P command cannot be used inside a stored form sequence. For automatic printing of stored forms, use the PA command.

Example:

This example prints two label sets of one label each, see the command below:

`P2,1` ↵

The principles for how counters are printed by this example, where the print command is `P2,2`.

Counter: 1	Label No. 1
Counter: 1	Label No. 2
Counter: 2	Label No. 3
Counter: 2	Label No. 4

PA - Print Automatic

Purpose: This command is used in a stored form sequence to automatically print the form as soon as all variable data has been supplied.

Syntax: $PAp_1[,p_2]$

Parameter Descriptions

Parameter	Description
p_1	Number of label sets (1-65535).
p_2	Number of copies of each label (1-64435) that is used in combination with counters to print multiple copies of the same label.

Remarks: Refer to the P command for explanations on how to print multiple labels with counters. The PA command follows the same principles.

Caution

The PA command can only be used with forms containing at least one variable (see V command). If there is no variable in the form, the printer will enter a loop and print continuously.

Example:

```
FK"TEST6" ↵
FS"TEST6" ↵
V00,59,N,"Enter text" ↵
A24,24,0,4,1,1,N,V00 ↵
PA1 ↵
FE ↵
FR"TEST6" ↵
? ↵
This is variable text
```

PF - Feed Media

Purpose: This command is used to feed the media forward or backward a given length.

Syntax: PFp_1[,p_2[,p_3]]

Parameter Descriptions

Parameter	Description
p_1	Number of steps to feed the media. Positive values will feed the media forward, negative values backward. Range is -999 to 9999.
p_2	Position control after feeding. Allowed values: 0 = The media has unknown position. Printer needs label gap/black mark detecting before next print job (default). 1 = Media is located in the printing position. 2 = Media feed will stop when Black mark/label gap is in sensor position or, if no black mark/gap is detected when p_1 dots are stepped (and error 99 will be reported).
p_3	Option delay after feeding. Delay will be p_3*100 ms.

Remarks: Media feeding is disrupted by pressing the **Feed** key or by opening the cover.

Example 1:

The following command will feed the media forward 360 steps, make the printer detect gap/black mark before the next print job, and cause a delay of 3 seconds after the feed. The red LED will blink during the delay.

```
PF360,0,30 ↵
```

Example 2:

The following command will retract the media 360 steps, and assume that the media is in printing position.

```
PF-360,1 ↵
```

Q - Set Form Length (Gap or Slot)

Purpose: This command is used to set the form and gap length when using the label gap sensor, or the amount of media feed after the print image in case of continuous stock.

Syntax: Qp1,p2[±p3]

Parameter Descriptions

Parameter	Description
p_1	Form length measured in dots. **203.2 dpi (8 dots/mm):** Default is 1200 dots. Maximum without memory cartridge is 3780 dots at full print width. Maximum with 256k SRAM cartridge is 6258 dots at full print width. **300 dpi (11.81 dots/mm):** Default is 1200 dots. Maximum without memory cartridge is 2640 dots at full print width. Maximum with 256k SRAM cartridge is 4364 dots at full print width.
p_2	Gap length measured in dots. 203.2 dpi (8 dots/mm) at 24 dots is the default. 300 dpi (11.81 dots/mm) at 24 dots is the default.
$\pm p_3$	Optional offset length measured in dots.

Remarks:

Gaps and Slots

The printers have a label gap sensor designed to detect the top of each form. It does this by:

- through the semi-transparent liner in the gap between labels.
- through a hole in the media.

The sensor is located slightly to the right in relation to the center of the media path (as seen from the printer's front). Refer to your printer's user's manual for specifications of the size and location of detection slots.

When entering the Dump mode (see Chapter 1), or when printing a form for the first time after power-up using the Windows Driver, the printer automatically determines the Q value while feeding a couple of labels. The current Q value is printed on the test label and the label produced by a U command.

Continuous Stock

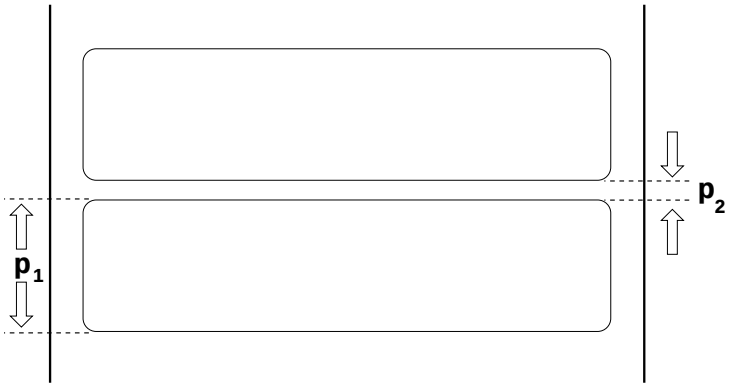
In case of continuous stock, parameter p_1 decides the amount of media feed performed after the actual print image has been printed. Continuous stock is selected by setting parameter $p_2 = 0$.

Be careful not having the printer loaded with continuous stock when entering the Dump mode or an error will occur.

Example 1:

Rectangular label (203.2 dpi = 8 dots/mm printhead):

p_1 = 20.0 mm (160 dots)
p_2 = 3.0 mm (24 dots)



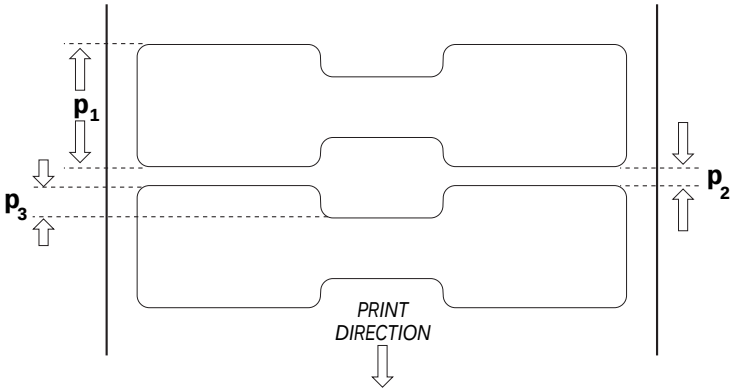
The Q command would be:

```
Q160,24 ↵
```

Example 2:

Butterfly label (293.2 dpi = 8 dots/mm printhead):

p_1 = 12.5 mm (100 dots)
p_2 = 3.0 mm (24 dots)
p_3 = 3.0 mm (24 dots)



The Q command would be:

```
Q100,24+24 ↵
```

Q - Set Form Length (Black Mark)

This command is used switch from label gap sensor to the black mark sensor, and to specify the location and height of the black marks on the back of the media.

Syntax: $Qp_1,Bp_2[\pm p_3]$

Parameter Descriptions

Parameter	Description
p_1	Distance between black marks measured in dots.
B	Disables label gap sensor and enables the black mark sensor.
p_2	Height of black mark measured in dots.
$\pm p_3$	Optional offset length measured in dots.

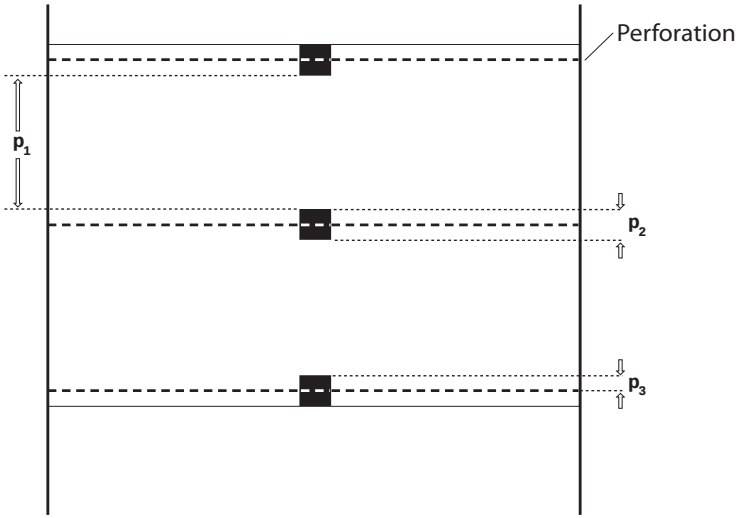
Remarks: In addition to the label gap sensor, the printers have a black mark sensor that determines the top of each form by sensing a preprinted black mark on the back of the media. The sensor is located slightly to the right in relation to the center of the media path (as seen from the printer's front).

Refer to your printer user's manual for specifications of the size and location of black marks.

Example 1:

On this tag, the black marks are printed on the perforation in a 203.2 dpi (8 dots/mm) printer:

p_1 = 31 mm (248 dots)
p_2 = 7.0 mm (56 dots)
p_3 = 0.5 mm (4 dots)



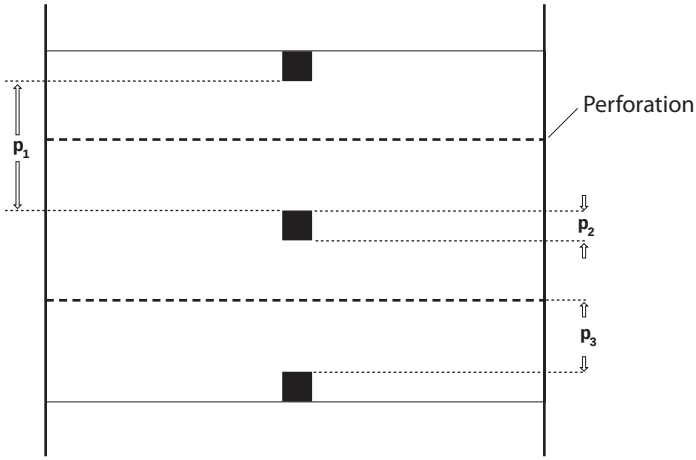
The Q command would be:

```
Q248,B56+4 ↵
```

Example 2:

On the tag below, the black marks are printed between the perforations. The printer has a 203.2 dpi (8 dots/mm) printhead.

p_1 = 31 mm (248 dots)
p_2 = 7.0 mm (56 dots)
p_3 = 17 mm (136 dots)



The Q command would be:

```
Q248,B56-136 ↵
```

q - Set Label Width

Purpose: This command is used to set the label width when using less than full width labels.

Syntax: qp_1

Parameter Descriptions

Parameter	Description
p_1	Width of label measured in dots. Default: 832 at 203.dpi (8 dots/mm) 1208 at 300 dpi (11.81 dots/mm)

Remarks: The q command will cause the image buffer to be formatted to match the label width, that is width is traded off for increased length within the same memory size.

The q command will also automatically set the margins according to the following rule:

```
(Number of dots on printhead - label width in dots)
/ 2 (center-aligned)
```

Note: If an R command (reference point) is sent after a q command, the image buffer will automatically be reformatted to match the width of the printhead and the margins will be reset accordingly.

Note: At 300 dpi, a maximum label of 1248 dots can be set. However, the print image will not be aligned at the center, and the data in the outer margins (beyond `q1232`) will be lost.

Example:

This example sets the label width to 416 dots.

`q416` ↵

R - Set Reference Point

Purpose: This command is used to move the reference point for the X and Y axis. All horizontal and vertical measurements in other commands use the setting for R as the origin for measurements.

Syntax: Rp_1,p_2

Parameter Descriptions

Parameter	Description
p_1	Horizontal (left) margin measured in dots. Default is 000.
p_2	Vertical (top) margin measured in dots. Default is 000.

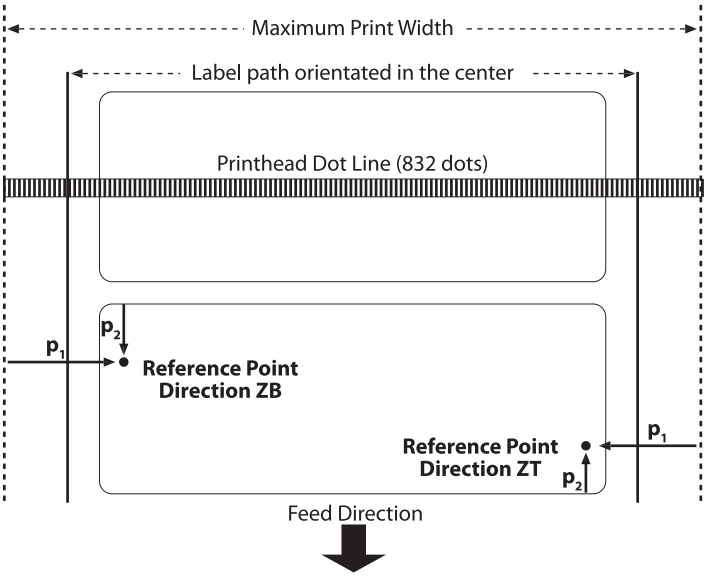
Remarks: The reference point command is used to establish top and left margins to prevent printing off the edge of the label. A minimum margin of 1 mm should be used on all sides of the label.

Caution

Repeated printing outside the edge of the media can cause excessive printhead wear.

For narrow labels, the R command could be substituted by q and r commands, which have the benefit of making better use of a limited image buffer. Any R command after a q command will revoke the width to the maximum print width, as a q832 command. The values for the latest R command (or r, whichever was last executed) are printed on the test label.

The print direction commands ZB, and ZT affect the location of the reference point, as illustrated below.



Example:

This example shows creates a 50 dot left margin on a label, see command below:

```
R50,100 ↵
```

r - Set Relative Reference Point

Purpose: This command is used to move the reference point for the x-axis and y-axis relative to the label width set by a previous q command.

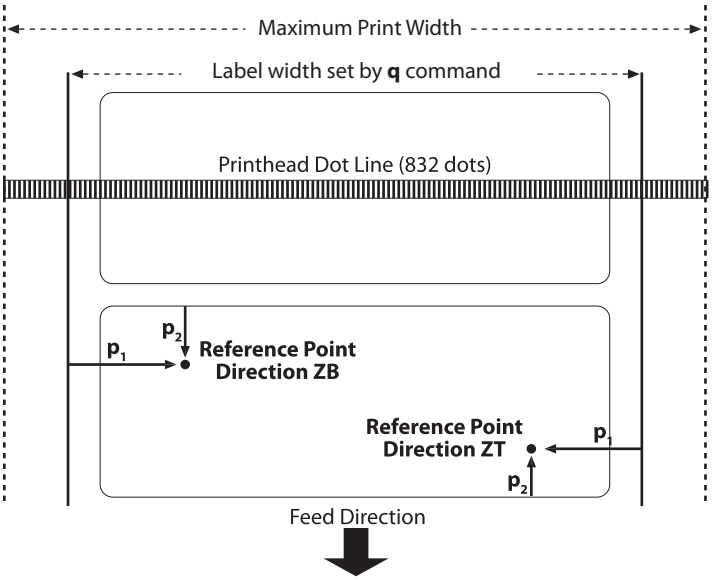
Syntax: rp_1,p_2

Parameter Descriptions

Parameter	Description
p_1	Horizontal (left) margin measured in dots. Default is 000.
p_2	Vertical (top) margin measured in dots. Default is 000.

Remarks: The relative reference point command is used to establish the top and left margins relative to the label width. For narrow labels, the r command can be used after a q command, which has the benefit of making better use of a limited image buffer. The values for the latest r command (or R, whichever was last executed) are printed on the test label.

The print direction commands ZB and ZT affect the location of the relative reference point, see the illustration on the next page.



Example:

This example creates a 50 dot left margin and a 100 dot top margin from the label edge set by q command.

```
r50,100 ↵
```

S - Speed Select

Purpose: This command is used to select the print speed.

Syntax: Sp_1

Parameter Descriptions

Parameter	Description
p_1	Speed select value: 0 = 30 mm/sec (1.2 in/sec). 1 = 40 mm/sec (1.6 in/sec). 2 = 50 mm/sec (2 in/sec). 3 = 75 mm/sec (3 in/sec). 4 = 100 mm/sec (4 in/sec). This speed is for 203.2 (8 dots/mm) printers only.

Remarks: Changing the print speed can affect the blackness of the printout, which may have to be adjusted using a D command. The density settings has priority over speed selection, which might cause the speed to be reduced at high density settings.

Example:

This example shows how to set the print speed to 500 mm/sec (2 in/sec).

`S2` ↵

SA - Setup Adjust

Purpose: This command is used to control the activation and storage of setup commands and values.

Syntax: SAp_1

Parameter Descriptions

Parameter	Description
p_1	Setup adjust value: 0 = All commands remain active and values are saved when changed. The printer will use stored setup values (default). 1 = All setup parameters are locked and commands will be ignored. The printer will use previously stored setup values. 2 = All setup commands within forms will be ignored.

Remarks: The SA command affects the following setup commands: R, r, D, j, J, q, Q, S, Z, and SPC. The current SA value is stored in Flash memory and printed on the test label.

Example:

This example shows how to ignore all setup commands within a form.

`SA2` ↵

SPC - Setup Print Copy

Purpose: This command is used to change the function of the **Feed** key to print a copy of the label residing in the printer's image buffer.

Syntax: SPCp_1

Parameter Descriptions

Parameter	Description
p_1	Setup Print Copy value: 0 = Media will feed as usual when the **Feed** key is pressed (default). 1 = Print Copy is activated, but will be disabled if the A, b, B, GW, GG, LE, LO, LS, LW, N, q, Q, r, R, X, and Info commands have been sent since the last printout. 2 - Print Copy is activated. In this state, additional fields can be added to the existing image and printed by pressing the **Feed** key.

Remarks: The following commands will disable Print Copy when SPC1 is set: A, b, B, GW, GG, LE, LO, LS, LW, N, q, Q, r, R, X, and Info. The current SPC value is stored in Flash memory and printed on the test label.

Example:

This example shows how to activate Print Copy, see command below:

```
SPC1 ↵
```

TD - Define Date Format

Purpose: This command is used to define the date format when printing.

Syntax: TDp_1[/p_2/p_3]

Parameter Descriptions

Parameter	Description
P1 - P3	The parameters describe the format of the date display. At least one parameter must be supplied. Each parameter can be any of the acceptable values listed below: y2 = Year displayed as 2 digits. y4 = Year displayed as 4 digits. me = Month displayed as a 3 letter English abbreviation. mn = Month displayed as 2 digits. dd = Day displayed as 2 digits. / = An optional separator character, which can be any character in the range between ASCII 32 dec. and ASCII 63 dec. The separator is printed between the results of each of the supplied parameters. Default is me/dd/y4.

Remarks: This command works only if the printer is fitted with an optional real-time clock circuit (RTC).

Example 1:

If the current date is January 15, 2004:

`TDy2/me/dd` ↵

The printer will print 04/JAN15.

Example 2:

If the current date is January 15, 2004:

`TDdd-me-y4` ↵

The printer will print 15-JAN-2004.

Example 3:

If the current date is January 15, 2004:

`TDme/dd/y4` ↵

The printer will print JAN/15/2004.

TS- Set Real-Time Clock

Purpose: This command is used to set the time and date in the printer's optional real-time clock circuit.

Syntax: TSp_1,p_2,p_3,p_4,p_5,p_6

Parameter Descriptions

Parameter	Description
p_1	Months are entered as 0-12.
p_2	Days are entered as 01-31.
p_3	Years are entered as the last two digits. For example 2004 will be 04.
p_4	Hours will be entered in 24 hour format (00-23).
p_5	Minutes are entered as 00-59.
p_6	Seconds are entered as 00-59.

Remarks: This command works only if the printer is fitted with an optional real-time clock circuit (RTC).

Example:

This example sets the date to JAN 15,2004 and the time to 12:45:23 p.m.

```
TS01,15,04,12,45,23 ↵
```

U- Print Configuration (General)

Purpose: This command is used to print the current printer configuration.

Syntax: U

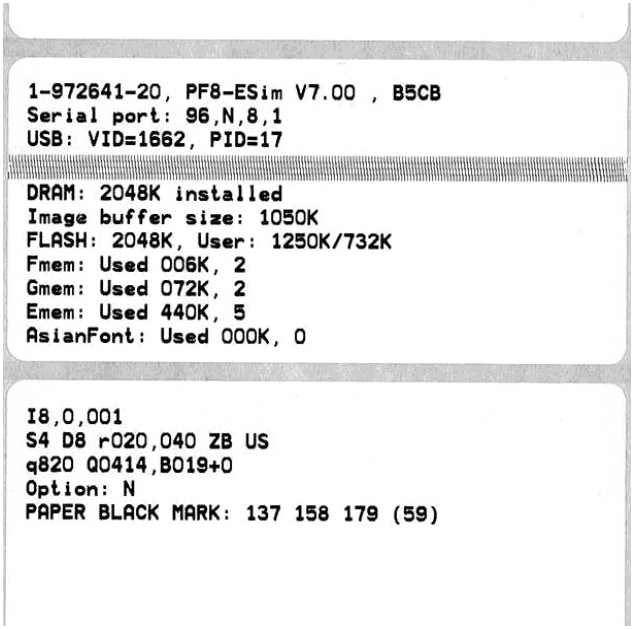
Remarks: This command produces a label with current printer configuration settings without entering Dump mode.

Example:

This example shows a typical label printed when using the U command with the following setup:

- Memory cartridge loaded
- Media with black marks
- Two forms, two graphics, and five soft fonts loaded

- Short labels



An example of a PF8's printer configuration label.

UC - Command Reply Accept Character Control

Purpose: This command specifies a prompt acknowledge character which the printer returns on the RS-232 or USB channel to the host after each command execution with no error reply before proceeding with the next command.

Syntax: UCp_1

Parameter Descriptions

Parameter	Description
p_1	Specifies the desired reply character by its ASCII decimal value (0-255). If p_1 is anything other than numerical characters or the value is 0, there will be not be any Command Reply Accept character (default).

Remarks: Default is no Command Reply Accept Character (command UC0).

Parameter p_1 will save in Flash.

If command UC is activated, it will replace the <ACK> character at the end of command US and P.

If an error occurs and error report is activated, the error message will be sent instead.↵

Example:

The table below shows an example of the UC command with the a reply.

Command	Reply	Explanation
`UC062` ↵	>	Set Command Reply Accept character to “>”.
`US` ↵	>	Reply error.
`X50,200,5,400,20` ↵	>	Draw a box.
`P` ↵	>	Print box and reply “>” at ready.

UDC - Memory Cartridge Information

Purpose: This command checks an inserted memory cartridge and returns information on its content on the RS-232 or USB channel.

Syntax: UDC

Remarks: This command checks the memory cartridge for the following information and returns one of the following alternatives listed in the table below.

Memory Cartridge Commands and Explanations

Command	Explanation
`NA` ↵	Cartridge not detected.
`000` ↵	Cartridge detected, but no relevant information is found.
`001-004` ↵	Cartridge detected. Followed by one to four lines of information on detected firmware, and/or Asian fonts.

Example:

The example below shows a UDC command and a sample response.

```
UDC ↵
004 ↵
1-972698-02,Cardfirm 02 ↵
1-972601-01,Base 4.01 ↵
1-972699-01,Bootfirm 01 ↵
1-972697-33,Intfont 01 ↵
```

UE - Soft Font Information Inquiry

Purpose: This command makes the printer send information back to the host on the soft fonts stored in memory.

Syntax: UE

Remarks: The printer sends the number of soft fonts and the name, height, and direction of each soft font through the RS0232 or USB port.

The UE command will be executed directly, without appending any Linefeed.

Example:

```
UE ↵
```

UF - Form Information Inquiry

Purpose: This command will cause the printer to send information about forms currently stored in the printer back to the host.

Syntax: UF

Remarks: The printer will send the number of forms stored and the name of each form to the host through the RS-232 or USB port.

The UF command will be executed directly, without appending any Linefeed.

UG - Graphics Information Inquiry

Purpose: This command will cause the printer to send information about graphics currently stored in the printer back to the host.

Syntax: UG

Remarks: The printer will send the number of forms stored and the name of each form to the host through the RS-232 or USB port.

The UG command will be executed directly, without appending any Linefeed.

UI - Enable Prompts/Code Page Inquiry

Purpose: This command will cause the printer to enable prompts to be sent to the host and to send the currently selected code page to the host through the RS-232 or USB port.

Syntax: UI

The printer will send information on the currently selected code page back to the host in the following formats:

UIp_1,p_2,p_3

Parameter Descriptions

Parameter	Description
p_1	Number of data bits.
p_2	Code page.
p_3	Country code.

Remarks: The keyboard display unit automatically sends this command each time power is applied. The UI command is disabled by removing power from the printer for 60 seconds.

Also see **"I - Character Selection" on page 87** and **"U- Print Configuration (General)" on page 117**.

Example:

This example enables prompts from the host and returns the current code page.

```
UI ↵
UI80,001
```

UM - Code Page and Memory

This command will cause the printer to send the currently selected code page and memory status to the host through the RS-232 or USB port.

Syntax: UM

The printer will send information on the currently selected code page and memory status back to the host in the following formats:

UMp_1,p_2,p_3,p_4,p_5,$p_6$$p_7$ UIp_8,p_9,p_{10}[,p_{11}]

Parameter Descriptions

Parameter	Description
p_1	Image buffer size in kilobytes.
p_2	Total user Flash memory size for [Form + Graphic + External font + Asian font] in kilobytes including decimals.
p_3	Form memory used in kilobytes. Decimals = Forms file counter.
p_4	Graphic memory allocation size in kilobytes.
p_5	Graphic file counter.
p_6	External font memory allocation size in kilobytes.
p_7	External font file counter.
p_8	Number of data bits.
p_9	Code page.
p_{10}	Country code.
p_{11}	Asian font file flag and used memory in kilobytes. Annn = Chinese traditional Bnnn = Chinese simplified Cnnn = Japanese Dnnn = Korean where nnn = memory used in kilobytes for Asian font.

Remarks: Also see "I - Character Selection" on page 87, "U- Print Configuration (General)" on page 117, "UI - Enable Prompts/ Code Page Inquiry" on page 121, and "UP - Code Page and Memory Inquiry Print" on page 123.

Example:

This example returns the current code page, see command below:

UM ↵

```
UM1050,998.0,8.3,40,1,37,2 UI80,001,B581
```

UN - Disable Error Reporting

Purpose: This command is used to disable error reporting.

Syntax: UN

Remarks: Cancels US command.

UP - Code Page and Memory Inquiry Print

Purpose: This command will cause the printer to print and send the currently selected code page, and memory status to the host through the RS-232 or USB port.

Syntax: UP

The printer will:

- send information on the currently selected code page and memory status back to the host (same as UM command).
- print the current printer configuration.

The format of the data sent to the host in the following formats:

UMp_1,p_2,p_3,p_4,p_5,p_6,p_7 UIp_8,p_9,p_{10}

Parameter Descriptions

Parameter	Description
p_1	Image buffer size in kilobytes.
p_2	Form memory allocation size in kilobytes.
p_3	Form memory free in kilobytes.
p_4	Graphic memory allocation size in kilobytes.
p_5	Graphic memory free in kilobytes
p_6	External font memory allocation size in kilobytes.
p_7	External font memory free in kilobytes.
p_8	Number of data bits.
p_9	Code page.
p_{10}	Country code.

Remarks: Also see "I - Character Selection" on page 87, "U- Print Configuration (General)" on page 117, and "UI - Enable Prompts/Code Page Inquiry" on page 121.

US - Enable Error Reporting

Purpose: This command is used to enable the printer's status reporting feature with optional direct recovery.

Syntax: US [A] [B] [E]

Parameter Descriptions

Parameter	Description
A	If any error of type A occurs, it will be directly recovered and continue in normal mode. No stop for recovery, often performed with the **Feed** key.
B	If any error of type A occurs, it will be directly recovered and continue in normal mode. No stop for recovery, often performed with the **Feed** key.
E	Error 13 LTS reply control (label taken sensor. Returns error 13 if LTS detects a label before and after printing.

Remarks: Any or all parameters can be activated at the same time. Refer to Appendix A for information on error types.

Parameters are saved in Flash memory.

The format of the error messages is controlled by the eR command.

Example:

The table on the next page shows an example of the US command with the a reply.

Command	Reply	Explanation
`USABE` ↵	<ACK>	Activate all error flags.
`eRQ,6,R` ↵		Setup user-defined error/status character control to Q for <XOFF> and R for <XON>.
`N` ↵		Clears image buffer errors.
`AA` ↵		Erroneous command type A (syntax error).
`P` ↵	<NAK>01Q	Print image buffer with error 01,
	R	Direct recovery error 01 (type A) and continue.

USR - Read Serial Number

Purpose: This command is used to read the serial number in EEPROM.

Note: This command can only be used on PF8 printers.

Syntax: USR

UV - Product Identity and Asian Font Types

Purpose: This command generates an output on the RS-232 or USB port about the software and font products stored in the printer's memory cartridge.

Syntax: UV

Remarks: The first line returned after the UV command is base software appended by a CR/LF, then comes one line with information on the font appended by CR/LF.

V - Define Variable

Purpose: This command is used to define variable data fields for use in stored forms.

Syntax: Vp_1,p_2,p_3,"PROMPT"

Parameter Descriptions

Parameter	Description
p_1	Variable reference number (00-99). A maximum total of 1500 bytes of data for all variables is allowed.
p_2	Maximum number of digits for the variable (1-99). A maximum total of 1500 bytes of data for all variables is allowed.
p_3	Field justification: L = Left justification. R = Right justification. C = Center justification. N = No justification.
[-]	A single leading minus sign in the prompt field will cause the prompt to be sent one time only after the form is retrieved (Keyboard Display Unit only).
[--]	A double leading minus sign in the prompt field will cause the prompt to be suppressed (Keyboard Display Unit only).
“PROMPT”	An ASCII text field that will be transmitted to the host or Keyboard Display Unit via the serial interface each time this command is executed. This prompt requests the operator to enter the value for the variable. Length limited to 32 characters.

Remarks: This command is used in forms that require unique data on each label. When initializing variables, they must be defined in order (V00, V01, V02, etc.) immediately after the FS command.

The field justification parameter affects the way the variable will be printed. When left, right, or center justification are selected, the counter value will be printed left, right or center justified in an area with a width defined by the p_2 parameter. If the number of digits in the counter value is less than the number of digits defined by p_2, the area will be padded with space characters.

If no justification is selected, the field will adjust to fit the actual length of the data and will not exceed the set maximum field length, which may be useful when using a counter as input data to a bar code.

To print the contents of a variable, the number of the variable must be included in the "DATA" field of the A (Print Text) or B (Print Bar Code) commands.

Example:

This example shows how the field justification works in variable fields.

```
FK"TEST7" ↵
FS"TEST7" ↵
V00,10,L,"Variable 00" ↵
V01,10,R,"Variable 01" ↵
V02,10,C,"Variable 02" ↵
V03,10,N,"Variable 03" ↵
A50,50,0,3,1,1,N,"TEXT"V00":Left justified" ↵
A50,100,0,3,1,1,N,"TEXT"V01":Right justified" ↵
A50,150,0,3,1,1,N,"TEXT"V02":Center justified" ↵
A50,200,0,3,1,1,N,"TEXT"V03":No justification" ↵
FE ↵
```

Refer to the ? command for a continuation of this example.

W - Windows Mode

Purpose: This command is used to enable/disable the Windows command mode (special applications only).

Syntax: Wp_1

Parameter Descriptions

Parameter	Description
p_1	Windows mode enable/disable: Y = Enables Windows mode. N = Disables Windows mode (default).

Remarks: When enabled, the printer will accept Windows mode escape sequences to print data. When disabled, escape sequences will be ignored up to the next linefeed.

The Windows mode escape sequences are only used by the Windows Printer Driver. When working with a mainframe or other non-Windows host, this mode can be disabled to prevent erratic operation.

X - Draw Box

Purpose: This command is used to draw a box shape.

Syntax: Xp_1,p2,p3,p4,p5

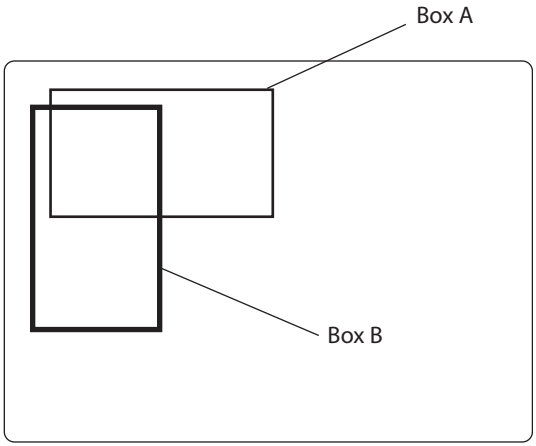
Parameter Descriptions

Parameter	Description
p_1	Horizontal start position (X) in dots.
p_2	Vertical start position (Y) in dots.
p_3	Line thickness in dots.
p_4	Horizontal end position (X) in dots.
p_5	Vertical end position (X) in dots.

Example:

This example draws two boxes (A and B) on a label, see commands below:

```
N ↵
X50,200,5,400,20 ↵
X200,50,10,20,400 ↵
P1 ↵
```



Y - Serial Port Setup

Purpose: This command is used to establish the communication parameters of the serial RS-232 port.

Syntax: Yp_1,p_2,p_3,p_4

Parameter Descriptions

Parameter	Description
p_1	Baud rate: 19 = 19,200 baud. 96 = 9,600 baud. 48 = 4,800 baud. 24 = 2,400 baud. 12 = 1,200 baud.
p_2	Parity: O = Odd. E = Even. N = None.

Parameter Descriptions (continued)

Parameter	Description
p_3	Number of data bits: 7 = 7 data bits. 8 = 8 data bits.
p_4	Number of stop bits: 1 = 1 stop bit. 2 = 2 stop bit.

Remarks: After receiving this command, the printer will automatically reset its communication on the serial RS-232 communication port. There are no such parameters for the USB port.

By default, the printer is set for 9600 baud, no parity, 8 data bits, 1 stop bit. XON/XOFF handshaking is always used. The printer sends XOFF when an error occurs. RTS/CTS is not supported.

If the current communication setup is not known, it can be checked by printing a test label, see **"Autosensing Mode" on page 3** for more information on how to print a test label.

Example:

In this example, we are setting a 19,200 baud rate, odd parity, 7 data bits, and 1 stop bit.

```
Y19,O,7,1 ↵
```

Z - Print Direction

Purpose: This command is used to select the print orientation

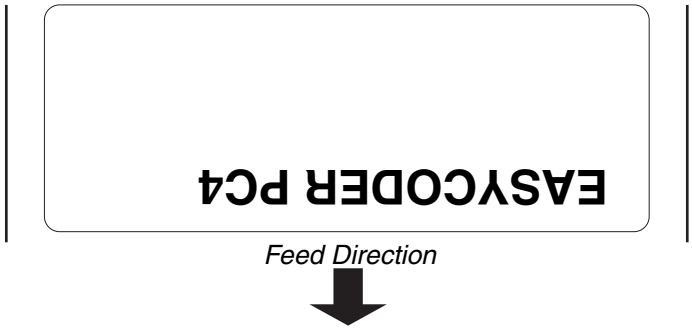
Syntax: Zp_1

Parameter Descriptions

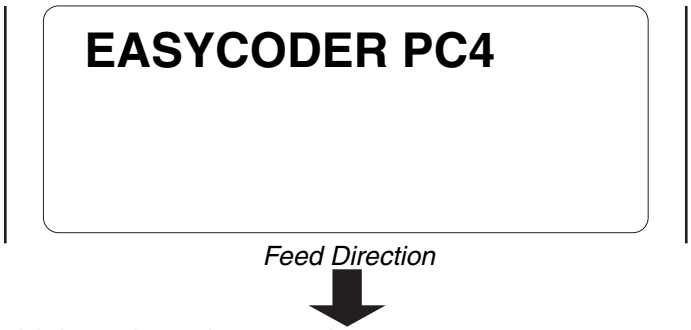
Parameter	Description
p_1	Print orientation: T = Start printing from the top of image buffer (default). B = Start printing from the bottom of image buffer.

Remarks: This command affects the complete print image, including text, bar codes, graphics, lines, and boxes, as well as the location f the reference point, see **"R - Set Reference Point" on page 110**.

Note: that printing a test label in the Test modes, or by using a U or UP command, will reset the print direction to default (ZT).



A label printed using the command ZT.



A label printed using the command ZB.

Example:

This example makes the printer print from the bottom of the image, see command below.

```
ZB ↵
```

? - Download Variables

Purpose: This command is used to signal to the printer that the data following are variable or counter values.

Syntax: ?

Remarks: This command is used by the host system to send data representing variables and/or counters to the printer after a stored for containing variables and/or counters has been retrieved. The amount of data following the question mark line must match exactly the total number and order of variables and/or counters for that specific form.

Note: If the ? command is omitted, no variables or counter values will be printed.

Example:

```
FR"TEST7" ↵
? ↵
12345 ↵
abcde ↵
ABCDE ↵
99999 ↵
P1 ↵
```

^@ - Reset Printer

Purpose: This command resets the printer.

Syntax: ^@

Remarks: The ^@ command resets the printer in the same way as a power off followed by a power on.

- the command must be followed by CR/LF to be correctly interpreted.
- the reset command is only available during command input, that is, not as data in PCX graphics, soft fonts, or in format forms.
- the printer will be unavailable for a short time during which all commands sent to the printer will be lost.

Note: An error report has higher priority than other commands and is processed directly, both in position of normal command input and in position of any error recovery.

An ^@ command cannot be performed anywhere else, such as:

- during downloading of images, fonts, or forms.
- inside a variable input (?) or inside a string area (“”).
- when the input data buffer is full (> 8000 bytes).

^A - Enter Pause Printing Mode

Purpose: This high-priority command makes the printer enter Pause Printing mode, that is, temporarily stop printing more labels as soon as the immediate label is completely printed and fed out (batch-printing only).

Syntax: ^A

Remarks: When the printer is in Pause mode, the control LED slowly blinks green, orange, and green again. The reply at command US is by default <NAK>92<XOFF> (can be changed using eR command). No reply at UN command.

To resume printing a batch, send the command ^C or press the **Feed** key.

To cancel the remainder of the print job, use the ^B command.

This command can only be sent via the RS-232 or USB ports.

Example:

This example renders a box and orders the printer to print 4 copies. At command ^A, the printing is paused and continued again at command ^C.

```
X0,0,10,20,40 ↵
P4 ↵
^A ↵
^C ↵
```

^B - Break Batch Printing

Purpose: This high-priority command makes the printer stop printing more labels as soon as the immediate label is completely printed and fed out and cancels the remainder of the print job (batch printing only). It also cancels print jobs that have already been paused using a ^A command.

Syntax: ^B

Remarks: The reply at US is by default <ACK> (can be changed using UCnn command).

This command can only be sent via RS-232 or USB ports.

Example:

This example renders a box and orders the printer to print 4 copies. At command ^B, the printing is stopped and cancelled.

```
X0,0,10,20,40 ↵
P4 ↵
^B ↵
```

^C - Exit Pause Printing Mode

Purpose: This high-priority command makes the printer exit the Pause Printing mode, that is, continue printing the remainder of a print job after having been temporarily stopped using a ^A command.

Syntax: ^C

Remarks: When the printer is in Pause Printing mode, the control LED slowly blinks green, yellow, and green again. The reply at command US is by default <XON> (can be changed using eR command). No reply at UN command.

Another way to resume printing a batch is to press the **Feed** key.

This command can only be sent via the RS-232 or USB ports.

Example:

This example renders a box and orders the printer to print 4 copies. At command ^A, the printing is paused and continued again at command ^C.

```
X,0,0,10,20,40 ↵
P4 ↵
^A ↵
```

```
^C ↵
```

^D - Remaining Labels

Purpose: This high priority command makes the printer return the number of labels remaining to be printed in a batch print job.

Syntax: ^D

Remarks: The reply at command US is by default Pnnnn<CR><LF>, where nnnn is the number of remaining labels.

This command can only be sent via the RS-232 or USB ports.

Example:

This example renders a box and orders the printer to print 4 copies. At command ^A, the printing is paused and interrogated for remaining number of labels. Printing is continued again at command ^C.

```
X0,0,10,20,40 ↵
P4 ↵
^A ↵
^D ↵
^C ↵
```

^default - Restore Factory Defaults

Purpose: This command will set all parameters back to the factory default.

Syntax: ^default

Remark: Many recommendations, especially when troubleshooting or updating firmware, require the need to start from a known state. This command returns the printer to its default setup. See **"Default Setup" on page 18** for the default value of setup parameters. A delay of two seconds should be observed before inputting other commands as the printer restarts after cleanup to factory default.

^ee - Immediate Error Report

Purpose: This command makes the printer report error and status immediately via the RS-232 or USB port.

Syntax: ^ee

Remarks: After the command is sent into printer via the RS-232 or USB interface, the printer returns an error and status report on serial port, when reaching a position of normal command-input or error-waiting-recovering.

Command must end with <CR> or <LF> to be correctly executed.

The printer will reply to the host via serial port in one of the following formats:

1. If no error code is available (normal case):

   `00<CR><LF>`

   00 = Status code OK

2. If only one error code is available:

   `XX<CR><LF>`

   XX = Error or status code

3. If more error or status codes are available:

   `XX,YY,ZZ..<CR><LF>`

   XX YY ZZ.. = Error or status codes.

Note: An error report has higher priority than other commands and is processed directly, both in position of normal command input and in position of any error recovery.

An ^ee command cannot be performed anywhere else, such as:

- during downloading of images, fonts, or forms.
- inside a variable input (?), or inside a string area (“”).
- when the input data buffer is full (> 8000 bytes).

At position error waiting recovering, the command ^ee must be sent into the printer via the RS-232 or USB interface to get an error report.

Example:

Command	Manual Action	Reply	Comments
US		<ACK>	Enables error reporting.
AA			Command with syntax error (01).
	Open cover	<NAK>11<XOFF>	Manual action to generate error (11) and wait in position of error waiting recovering.
UV			Command to print product ID will be queued.
^ee		01,11<CR><LF>	Command reports direct all error/ status.
	Close cover	<XON>1-972600-33, → → Base3.33<CR><LF>	Manual action to release error 11. Command **UV** will now be performed.
P		<NAK>01<XOFF>	Command to print, reports syntax error (01).
	Press **Feed** key	<XON>	<Manual action to release error status.

8 Fonts

This chapter lists the printer's resident fonts and illustrates the various font sizes.

Resident Fonts

Your printers support uppercase and lowercase characters for font sizes 1-4 and uppercase characters for font size 5. All fonts are non-proportional. The ASCII value of the different characters is determined by the **I** command setting.

203.2 dpi (8 dots/mm) (Illustrated in Chapter 8 and 9)

Font	Size (dots)	Size (points)	Characters/inch
1	8 x 12	6	20.3
2	10 x 16	7	16.9
3	12 x 20	10	14.5
4	14 x 24	12	12.7
5	32 x 48	24	5.6

300 dpi (8 dots/mm) (Not Illustrated)

Font	Size (dots)	Size (points)	Characters/inch
1	12 x 20	4	25
2	16 x 28	6	18.75
3	20 x 36	8	15
4	24 x 44	10	12.5
5	48 x 80	214	6.25

Font Sizes 1-5

Illustrated below are the various fonts that are illustrated in real size printed on an 8 dots/mm (203.2 dpi) and a 11.81 mm (300 dpi) printer.

9 Code Pages

This chapter contains tables of the characters available in the various code pages for the printer's resident fonts, see **"I - Character Selection" on page 87**.

When using 7-bit communication, only the characters between ASCII 0 dec. and ASCII 127 dec. can be used.

The characters available, when font size 5 is selected, see **"A - Printing Text" on page 41**, are indicated by a gray background in the tables.

For examples of the actual appearance of the characters, see **"Fonts" on page 139**.

For information and printout samples on Asian fonts, please refer to your user's manual.

Printout Samples

Code Page DOS 437 English (UK); Cmd I8,0,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	Ç	ü	é	â	ä	à	å	ç	ê	ë	è	ï	î	ì	Ä	Å
144 -	É	æ	Æ	ô	ö	ò	û	ù	ÿ	Ö	Ü	¢	£			ƒ
160 -	á	í	ó	ú	ñ	Ñ	a	º	¿			½	¼	¡		
176 -																
192 -																
208 -																
224 -		ß					µ									
240 -									°							

Code Page DOS 850 Multilingual (Latin 1); Cmd I8,1,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	Ç	ü	é	â	ä	à	å	ç	ê	ë	è	ï	î	ì	Ä	Å
144 -	É	æ	Æ	ô	ö	ò	û	ù	ÿ	Ö	Ü	ø	£			ƒ
160 -	á	í	ó	ú	ñ	Ñ	a	º	¿			½	¼	¡		
176 -						Á	Â	À						¢		
192 -							ã	Ã								
208 -			Ê	Ë	È		Í	Î	Ï						Ì	
224 -	Ó	ß	Ô	Ò	õ	Õ	µ			Ú		Ù				
240 -			=	¾	¶	§			°							

Code Page DOS 852 Latin 2 (Cyrillic II/Slavic); Cmd I8,2,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	Ç	ü	é	â	ä	ů	ć	ç	ł	ë	Ő	ő	î	Ź	Ä	Ć
144 -	É	Ĺ	Í	ô	ö	Ľ	ľ	Ś	ś	Ö	Ü	Ť	ť	Ł	×	č
160 -	á	í	ó	ú	Ą	ą	Ž	ž	Ę	ę		ź	Č	ş		
176 -						Á	Â	Ě	Ş					Ż	ż	
192 -							Ă	ă								¤
208 -	đ	Đ	Ď	Ë	ď	Ň	Í	Î	ě					Ţ	Ů	
224 -	Ó	ß	Ô	Ń	ń	ň	Š	š	Ŕ	Ú	ŕ	Ű	ý	Ý	ţ	
240 -						§						ű	Ř	ř		

Code Page DOS 860 Portuguese; Cmd I8,3,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	Ç	ü	é	â	ã	à	Á	ç	ê	Ê	è	Í	Ô	ì	Ã	Â
144 -	É	À	È	ô	õ	ò	Ú	ù	Ì	Õ	Ü	¢	£	Ù		Ó
160 -	á	í	ó	ú	ñ	Ñ	a	º	¿	Ò		½	¼	¡		
176 -																
192 -																
208 -																
224 -	α	ß	Γ	π	Σ	σ	µ	τ	Φ	Θ	Ω	δ		φ	ε	
240 -									°					²		

Code Page DOS 863 Canada (French);Cmd I8,4,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	Ç	ü	é	â	Â	à	¶	ç	ê	ë	è	ï	î	ì	À	§
144 -	É	È	Ê	ô	Ë	Ï	û	ù	¤	Ô	Ü	¢	£	Ù	Û	ƒ
160 -	|		ó	ú			³		Î			½	¼	¾		
176 -																
192 -																
208 -																
224 -	α	ß	Γ	π	Σ	σ	µ	τ	Φ	Θ	Ω	δ		ϕ	ε	
240 -									°					²		

Code Page DOS 865 Norway; Cmd I8,5,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	Ç	ü	é	â	ä	à	å	ç	ê	ë	è	ï	î	ì	Ä	Å
144 -	É	æ	Æ	ô	ö	ò	û	ù	ÿ	Ö	Ü	ø	£	Ø		ƒ
160 -	á	í	ó	ú	ñ	Ñ	a	o	¿			½	¼	¡		
176 -																
192 -																
208 -																
224 -	α	ß	Γ	π	Σ	σ	µ	τ	Φ	Θ	Ω	δ		ϕ	ε	
240 -									°					²		

Code Page DOS 857 Turkish; Cmd I8,6,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	Ç	ü	é	â	ä	à	å	ç	ê	ë	è	ï	î	ı	Ä	Å
144 -	É	æ	Æ	ô	ö	ò	û	ù	İ	Ö	Ü	ø	£	Ø	Ş	ş
160 -	á	í	ó	ú	ñ	Ñ	Ğ	ğ	¿	®		½	¼	¡		
176 -						Á	Â	À	©					¢	¥	
192 -							ã	Ã								¤
208 -	o	a	Ê	Ë	È		Í	Î	Ï					|	Ì	
224 -	Ó	ß	Ô	Ò	õ	Õ	µ		χ	Ú	Û	Ù	ì	ÿ		
240 -					¶	§			°			¹	³	²		

Code Page DOS 861 Icelandic; Cmd I8,7,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	Ç	ü	é	â	ä	à	å	ç	ê	ë	è	Ð	ð	Þ	Ä	Å
144 -	É	æ	Æ	ô	ö	þ	û	Ý	ý	Ö	Ü	ø	£			ƒ
160 -	á	í	ó	ú	Á	Í	Ó	Ú	¿			½	¼	¡		
176 -																
192 -																
208 -																
224 -	α	ß	Γ	π	Σ	σ	µ	τ	Φ	Θ	Ω	δ		φ	ε	
240 -									°					²		

Code Page DOS 862 Hebrew; Cmd I8,8,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	א	ב	ג	ד	ה	ו	ז	ח	ט	י	ך	כ	ל	ם	מ	ן
144 -	נ	ס	ע	ף	פ	ץ	צ	ק	ר	ש	ת	¢	£	¥		ƒ
160 -	á	í	ó	ú	ñ	Ñ	a	º	¿			½	¼	¡		
176 -																
192 -																
208 -																
224 -	α	ß	Γ	π	Σ	σ	µ	τ	Φ	Θ	Ω	δ		φ	ε	
240 -									°					²		

Code Page DOS 855 Cyrillic; Cmd I8,9,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	ђ	Ђ	ѓ	Ѓ	ё	Ё	є	Є	s	S	i	І	ї	Ї	j	J
144 -	љ	Љ	њ	Њ	ћ	Ћ	ќ	Ќ	ў	Ў	ц	Ц	ю	Ю	ъ	Ъ
160 -	а		б	Б	ц	Ц	д	Д	е	Е	ф	Ф	г	Г		
176 -						х	Х	и	И					й	Й	
192 -							к	К								¤
208 -	л	Л	м	М	н	Н	о	О	п					П	я	
224 -	Я	р	Р	с	С	т	Т	у	У	ж	Ж	в	В	ь	Ь	№
240 -		ы	Ы	з	З	ш	Ш	э	Э	щ	Щ	ч	Ч			

Code Page DOS 866 Cyrillic; Cmd I8,10,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	А	Б	В	Г	Д	Е	Ж	З	И	Й	К	Л	М	Н	О	П
144 -	Р	С	Т	У	Ф	Х	Ц	Ч	Ш	Щ	Ъ	Ы	Ь	Э	Ю	Я
160 -	а	б	в	г	д	е	ж	з	и	й	к	л	м	н	о	п
176 -																
192 -																
208 -																
224 -	р	с	т	у	ф	х	ц	ч	ш	щ	ъ	ы	ь	э	ю	я
240 -	Ё	ё	Є	є	Ї	ї	Ў	ў	°				№	¤		

Code Page DOS 737 Greek; Cmd I8,11,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	Α	Β	Γ	Δ	Ε	Ζ	Η	Θ	Ι	Κ	Λ	Μ	Ν	Ξ	Ο	Π
144 -	Ρ	Σ	Τ	Υ	Φ	Χ	Ψ	Ω	α	β	γ	δ	ε	ζ	η	θ
160 -	ι	κ	λ	μ	ν	ξ	ο	π	ρ	σ	ς	τ	υ	φ	χ	ψ
176 -																
192 -																
208 -																
224 -	ω	ά	έ	ή	ϊ	ί	ό	ύ	ϋ	ώ	Ά	Έ	Ή	Ί	Ό	Ύ
240 -	Ώ				Ϊ	Ϋ			°					²		

Code Page DOS 841 Greek 1; Cmd I8,12,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	Ç	ü	é	â	ä	à	Ά	ç	ê	ë	è	ï	î	Έ	Ä	Ή
144 -	Ί		Ό	ô	ö	Ύ	û	ù	Ώ	Ö	Ü	ά	£	έ	ή	ί
160 -	ϊ	ΐ	ό	ύ	Α	Β	Γ	Δ	Ε	Ζ	Η	½	Θ	Ι		
176 -				ϕ		Κ	Λ	Μ	Ν					Ξ	Ο	
192 -							Π	Ρ								Σ
208 -	Τ	Υ	Φ	Χ	Ψ	Ω	α	β	γ					δ	ε	
224 -	ζ	η	θ	ι	κ	λ	μ	ν	ξ	ο	π	ρ	σ	ς	τ	
240 -			υ		χ	$	ψ		°		ω	ϋ	ΰ	ώ		

Code Page DOS 869 Greek 2; Cmd I8,13,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -							Ά				|			Έ		Ή
144 -	Ί	Ϊ	Ό			Ύ	Ϋ	©	Ώ	²	³	ά	£	έ	ή	ί
160 -	ϊ	ΐ	ό	ύ	Α	Β	Γ	Δ	Ε	Ζ	Η	½	Θ	Ι		
176 -				ϕ		Κ	Λ	Μ	Ν					Ξ	Ο	
192 -							Π	Ρ								Σ
208 -	Τ	Υ	Φ	Χ	Ψ	Ω	α	β	γ					δ	ε	
224 -	ζ	η	θ	ι	κ	λ	μ	ν	ξ	ο	π	ρ	σ	ς	τ	
240 -			υ		χ	§	ψ		°		ω	ϋ	ΰ	ώ		

Code Page Windows 1252 Latin 1; Cmd I8,A,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}	Ž	
128 -	€		‚								Š	‹	š	Ť	Ž	Ź
144 -	Ÿ										Ś	›	ś	ť	ž	ź
160 -				Ł	¤	Ą	|	§		©	Ş				®	Ż
176 -	°			ł		µ	¶			ą	ş		Ľ		ľ	ż
192 -	Ŕ	Á	Â	Ă	Ä	Ĺ	Æ	Ç	È	É	Ê	Ë	Ì	Í	Î	Ď
208 -	Đ	Ñ	Ò	Ó	Ô	Õ	Ö	χ	Ø	Ù	Ú	Û	Ü	Ý	Ţ	ß
224 -	ŕ	á	â	ă	ä	ĺ	ć	ç	č	é	ę	ë	ě	í	î	ď
240 -	đ	ń	ň	ó	ô	ő	ö		ř	ů	ú	ű	ü	ý	ţ	

Code Page Windows 1250 Latin 2; Cmd I8,B,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}	Ž	
128 -	€		‚								Š	‹	š	Ť	Ž	Ź
144 -	Ÿ										Ś	›	ś	ť	ž	ź
160 -				Ł	¤	Ą	|	§		©	Ş				®	Ż
176 -	°			ł		µ	¶			ą	ş		Ľ		ľ	ż
192 -	Ŕ	Á	Â	Ă	Ä	Ĺ	Æ	Ç	È	É	Ê	Ë	Ì	Í	Î	Ď
208 -	Đ	Ñ	Ò	Ó	Ô	Õ	Ö	χ	Ø	Ù	Ú	Û	Ü	Ý	Ţ	ß
224 -	ŕ	á	â	ă	ä	ĺ	ć	ç	č	é	ę	ë	ě	í	î	ď
240 -	đ	ń	ň	ó	ô	ő	ö		ř	ů	ú	ű	ü	ý	ţ	

Code Page Windows 1251 Cyrillic; Cmd I8,C,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	Ђ	Ѓ	‚	ѓ							Љ	‹	Њ	Ќ	Ћ	Џ
144 -	ђ										љ	›	њ	ќ	ћ	џ
160 -		Ў	ў	J	¤	Ґ	|	§	Ё	©	Є				®	Ї
176 -	°		I	i	ґ	µ	¶		ё	№	є		j	S	s	ї
192 -	А	Б	В	Г	Д	Е	Ж	З	И	Й	К	Л	М	Н	О	П
208 -	Р	С	Т	У	Ф	Х	Ц	Ч	Ш	Щ	Ъ	Ы	Ь	Э	Ю	Я
224 -	а	б	в	г	д	е	ж	з	и	й	к	л	м	н	о	п
240 -	р	с	т	у	ф	х	ц	ч	ш	щ	ъ	ы	ь	э	ю	я

Code Page Windows 1253 Greek; Cmd I8,D,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	€		‚	ƒ								‹				
144 -							–					›				
160 -				Ά	£	¤	¥	|	§		©	a			®	
176 -	°		²	³		µ	¶		Έ	Ή	Ί		Ό	½	Ύ	Ώ
192 -	ΐ	Α	Β	Γ	Δ	Ε	Ζ	Η	Θ	Ι	Κ	Λ	Μ	Ν	Ξ	Ο
208 -	Π	Ρ		Σ	Τ	Υ	Φ	Χ	Ψ	Ω	Ϊ	Ϋ	ά	έ	ή	ί
224 -	ΰ	α	β	γ	δ	ε	ζ	η	θ	ι	κ	λ	μ	ν	ξ	ο
240 -	π	ρ	ς		τ	υ	φ	χ	ψ	ω	ϊ	ϋ	ό	ύ	ώ	

Code Page Windows 1254 Turkish; Cmd I8,E,001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	€		,	ƒ					ˆ		Š	‹	Œ			
144 -											š	›	œ			Ÿ
160 -		¡	¢	£	¤	¥	|	§		©	a				®	
176 -	°		²	³		µ	¶			¹	o		¼	½	¾	¿
192 -	À	Á	Â	Ã	Ä	Å	Æ	Ç	È	É	Ê	Ë	Ì	Í	Î	Ï
208 -	Ğ	Ñ	Ò	Ó	Ô	Õ	Ö	χ	Ø	Ù	Ú	Û	Ü	İ	Ş	ß
224 -	à	á	â	ã	ä	å	æ	ç	è	é	ê	ë	ì	í	î	ï
240 -	ğ	ñ	ò	ó	ô	õ	ö		ø	ù	ú	û	ü	ı	ş	ÿ

Code Page Windows 1255 Hebrew; Cmd I8,F001

16 -					¶	§										
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}		
128 -	€		,	ƒ					ˆ			‹				
144 -												›				
160 -		¢	£		¥	|	§			©					®	
176 -	°		²	³		µ	¶			¹			¼	½	¾	
192 -																
208 -																
224 -	א	ב	ג	ד	ה	ו	ז	ח	ט	י	ך	כ	ל	ם	מ	ן
240 -	נ	ס	ע	ף	פ	ץ	צ	ק	ר	ש	ת					

Code Page In Dump Mode

0 -	00	☺	☻	♥	♦	♣	♠	•	◘	○	◙	♂	♀	♪	♫	☼
16 -	►	◄	↕	‼	¶	§	▬	↨	↑	↓	→	←	∟	↔	▲	▼
32 -		!	"	#	$	%	&	‘	(	)	*	+	,	-	.	/
48 -	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
64 -	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
80 -	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
96 -	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
112 -	p	q	r	s	t	u	v	w	x	y	z	{	|	}	~	⌂
128 -	Ç	ü	é	â	ä	à	å	ç	ê	ë	è	ï	î	ì	Ä	Å
144 -	É	æ	Æ	ô	ö	ò	û	ù	ÿ	Ö	Ü	ø	£	Ø	×	ƒ
160 -	á	í	ó	ú	ñ	Ñ	a	o	¿	®	¬	½	¼	¡	«	»
176 -	░	▒	▓	│	┤	Á	Â	À	©	╣	║	╗	╝	¢	¥	┐
192 -	└	┴	┬	├	─	┼	ã	Ã	╚	╔	╩	╦	╠	═	╬	¤
208 -	ð	Ð	Ê	Ë	È	ı	Í	Î	Ï	┘	┌	█	▄	¦	Ì	▀
224 -	Ó	β	Ô	Ò	õ	Õ	µ	þ	Þ	Ú	Û	Ù	ý	Ý	¯	’
240 -	-	±	=	¾	¶	§	÷	´	°	¨	˙	¹	³	²	■	255

A Error-Handling

This appendix explains the principles of error handling and lists the various error types and all possible error replies that the printer can return to the host via an RS-232 or USB channel. It also shows how the LED control lamp will switch between various colors to notify the operator of an error condition.

Error-Handling Principles

The firmware is able to handle a number of error conditions. Code numbers represent the different error cases.

The printer controls output of the error code and its format and waits until the error recovery is performed for each special error-code.

Error Action Source

A - Command Errors

The biggest group of error codes is command format and data information error. It exists errors of type “Syntax error” to more sophisticated like “Form or image name duplicate” (see error type A.).

B - Printing Error

Print out error occurs, when the printing the label not is correct performed, such as “Out of media,” “cover opened,” and “cutting not completed” (see error type B, C, and D).

C - Status

A status error performs normally not a true error case, but in some matter it has to be detected. A typical case is “LTS-detection waiting” (see error type E).

D - Special Mode

Some special feature errors will hang the printer forever or have a complete other output.

Typical of this group errors are: “Flashing not completed,” or “Dump mode” (see error type F and G.)

Error Action Output Information

The error code will often be written in the upper left corner of the label, “ERR01” for example. The control LED will change to orange or red.

Serial Port (RS-232 or USB)

Output on serial line when error occurs is dependent on some setup commands.

The command UN and US turn off or on most of the error output information.

The command eR specifies the format of the error output information.

Parallel Port

At an error, the parallel port will mark error status by setting pin 12 in active state.

Error Recover/Output

The normal action at any error is to stop in a waiting loop to recover the error feature. At this “Error Waiting Loop” only command ^ee and ^@ can be performed and executed.

The error recovery, often performed by pressing the **Feed** key, releases the error and sends recovery information on the serial line (RS-232 or USB), normally <XON>.

See also commands USA or USB for direct recovery.

Error-Handling Commands

This table shows the the error-handling commands and its descriptions.

Error-Handling Commands

Command	Description
US[A][B][E]	Optional error reporting.
UN	No error reporting.
eR	User defined error/status character control.
^ee	Error report - immediate
^@	Reset

Error Types

Classified types of error and status codes for the Error Replies table as seen on the next page.

Classified Types of Error and Status Codes

Type	Feature	Generates	Recovery
A	Minor error of command sequence and similar.	At error reporting (command US): Sets error on print command and sets the orange or red LED. At no error reporting (command UN): Green LED at print command	At Error reporting (command US): **Feed** key or command Reset **^@**. See also command USA. At no Error reporting (command UN): No recover necessary, already done.
B	Print stopping error.	Sets orange or red LED.	**Feed** key or command Reset **^@**. See also command USB.
C	Paper end stopping error.	Sets the red LED at print.	Load new media and press the **Feed** key or command reset **^@**.
D	Error of printing functionality and status.	Sets the red LED direct, sometimes blinking.	Manual or automatic (temperature lowered) measure or command reset **^@**.
E	Status changes.	No LED change.	No recover necessary/Delete Label from LTS. See also command reset **^@**.
F	Unrecoverable Error.	Sets red LED direct.	Power Off/ON or command reset **^@**.
G	Dump mode.	Green LED in Dump mode.	**Feed** key or command reset ^@
H	Pause mode	LED blinks green and orange.	**Feed** key or command ^B or ^C.

Error Replies

This table shows you an explanation of text for the Error Replies table on page "Error Replies Table" on page 158.

Error Text Description

Text	Description
Error	Referred error number divided in code and type.
Code	Referred error number.
Type	Letter is a classifying of error.
Description	Short human explanation.
Error response	Response depending on error reporting flag controlled by US and UN commands.
LED	Color of LED indicator: Green (GRN) Red (RED) Blinking red (BLNK RED) Orange (ORA)
Serial	Output on serial line (RS-232 or USB) at error response.
Response at ^ee	Output on serial line of error number, if only one error is available.
<LF>	0x0a
<CR>	0x0d
<CRLF>	0x0d + 0x0a
<XON>	0x11
<XOFF>	0x13
<NAK>	0x15
-	Nothing happens
[XOFF]	Flag for waiting on error recovery. Default value is <XOFF> = 0x13, but can be supstituted with command $eRp_1,6,p_3$ where p_1 is replacement of <XOFF> and where p_3 is a replacement of <XON>.

Example:

Command eRF,6,N substitutes at error reply <XOFF> with the character "F," and at error recovery <XON> with the character "N."

Error Replies Table

This table shows the Error response and codes reporting when Command US is on.

Error Response and Codes Reporting When Command US Is On

Error Code	Error Type	Description	LED	Serial	Response at Command ^ee
		No error	GRN		
01	A	Syntax error	RED	<NAK>01[XOFF]	01<CRLF>
02	B	Object exceeds image buffer border	ORA	<NAK>02[XOFF]	02<CRLF>
03	A	Data length error	RED	<NAK>03[XOFF]	03<CRLF>
04	A	Insufficient memory to store data	RED	<NAK>04[XOFF]	04<CRLF>
05	A	Memory configuration error	RED	<NAK>05[XOFF]	05<CRLF>
07	C	Ouyt of media, marked with Pnnn = Number of remaining labels.	RED	<NAK>07[XOFF]	07<CRLF>
08	A	Form or image name duplicate.	RED	<NAK>08[XOFF]	08<CRLF>
09	A	Form or image not found	RED	<NAK>09[XOFF]	09<CRLF>
11	D	Printhead up (cover open)	RED	<NAK>11[XOFF]	11<CRLF>
13	E	LTS detection waiting, peel pause.	-	<NAK>13[XOFF]	13<CRLF>
16	A	No form was retrieved before "?<CRLF>" was entered.	RED	<NAK>16[XOFF]	16<CRLF>

Error Response and Codes Reporting When Command US Is On (continued)

Error Code	Error Type	Description	LED	Serial	Response at Command ^ee
17	C	Out of ribbon, marked with Rnnn = Number of remaining labels.	RED	<NAK>07Rnnn [XOFF]	17<CRLF>
50	A	Does not fit in area specified.	RED	<NAK>50[XOFF]	50<CRLF>
51	A	Data length too long	RED	<NAK>51[XOFF]	51<CRLF>
61	D	High motor temperature	BLNK RED	<NAK>61[XOFF]	61<CRLF>
62	D	High printhead temperature	BLNK RED	<NAK>62[XOFF]	62<CRLF>
71	F	Wait after default setup.	GRN	-	71<CRLF>
72	F	Flashing not completed	RED	<NAK>72[XOFF]	72<CRLF>
73	F	Download error	RED	<NAK>73[XOFF]	73<CRLF>
81	B	Cutter jammed or not installed.	BLNK RED	<NAK>81[XOFF]	81<CRLF>
89	G	Dump mode (after auto-detection).	GRN	<NAK>89[XOFF]	89<CRLF>
92	H	Pause printing mode	BLNK GRN/ORA	<NAK>92[XOFF]	92<CRLF>
94	G	Autosensing mode	BLACK	<NAK>94[XOFF]	94<CRLF>
98	G	Download mode (not public use)			
99		Other errors		<NAK>99[XOFF]	99<CRLF>

This table shows the Error response and codes reporting when Command US is off.

Error Response and Codes Reporting When Command US Is Off

Error Code	Error Type	Description	LED	Serial	Response at Command ^ee
		No error	GRN	-	
01	A	Syntax error	GRN	-	01<CRLF>
02	B	Object exceeds image buffer border	ORA	[XOFF]	02<CRLF>
03	A	Data length error	GRN		03<CRLF>
04	A	Insufficient memory to store data	GRN	-	04<CRLF>
05	A	Memory configuration error	GRN	-	05<CRLF>
07	C	Ouyt of media, marked with Pnnn = Number of remaining labels.	RED	[XOFF]	07<CRLF>
08	A	Form or image name duplicate.	GRN	-	08<CRLF>
09	A	Form or image not found	GRN	-	09<CRLF>
11	D	Printhead up (cover open)	RED	[XOFF]	11<CRLF>
13	E	LTS detection waiting, peel pause.	-	-	13<CRLF>
16	A	No form was retrieved before “?<CRLF>” was entered.	GRN	-	16<CRLF>
17	C	Out of ribbon, marked with Rnnn = Number of remaining labels.	RED	[XOFF]	17<CRLF>
50	A	Does not fit in area specified.	GRN	-	50<CRLF>
51	A	Data length too long	GRN	-	51<CRLF>
61	D	High motor temperature	BLNK RED	[XOFF]	61<CRLF>

Error Response and Codes Reporting When Command US Is Off (continued)

Error Code	Error Type	Description	LED	Serial	Response at Command ^ee
62	D	High printhead temperature	BLNK RED	[XOFF]	62<CRLF>
71	F	Wait after default setup.	GRN	[XOFF]	71<CRLF>
72	F	Flashing not completed	RED	[XOFF]	72<CRLF>
73	F	Download error	RED	[XOFF]	73<CRLF>
81	B	Cutter jammed or not installed.	BLNK RED	[XOFF]	81<CRLF>
89	G	Dump mode (after auto-detection).	GRN	-	89<CRLF>
92	H	Pause printing mode	BLNK GRN/ORA	-	92<CRLF>
94	G	Autosensing mode	BLACK	[XOFF]	94<CRLF>
98	G	Download mode (not public use)			
99		Other errors		[XOFF]	99<CRLF>

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